

LI.
What are numbers and what should they be?
[First edition 1888. Sixth edition 1930.]

Ἀεὶ ὁ ἄνθρωπος ἀριθμητίζει.

Dedicated in heartfelt love
to my sister

Julie
and my brother
Adolf,

Doctor of Law, Higher Regional Court Councillor in Braunschweig.

Preface to the first edition

What is provable should not be believed in science without proof. However evident this demand appears, it is nevertheless, as I believe, still by no means fulfilled even in the foundation of the simplest science, namely of that part of logic which treats the doctrine of numbers, even after the most recent presentations.¹ In calling arithmetic (algebra, analysis) only a part of logic, I already express that I hold the concept of number to be entirely independent of the ideas or intuitions of space and time, and rather to be an immediate outflow of the pure laws of thought. My main answer to the question posed in the title of this writing is: numbers are free creations of the human mind; they serve as a means for apprehending the diversity of things more easily and more sharply. Through the purely logical construction of the science of numbers and through the continuous realm of numbers gained in it, we are first put in a position to investigate our ideas of space and time exactly, by relating them to this realm of numbers created in our mind.² If one follows precisely what we do in counting a multitude or number of things, one is led to consider the mind's capacity to relate things to things, to let one thing correspond to a thing, or to map a thing by means of a thing, without which capacity no thinking at all is possible. In my view the entire science of numbers must be erected on this single foundation, indispensable also in every other respect, as I already stated in an announcement of the present writing.³ I had already formed the intention of such a presentation before the publication of my writing on continuity; but only after its appearance, and with many interruptions caused by increased official duties and other necessary work, did I write down, in the years 1872 to 1878, a first draft on a few sheets. Several mathematicians then saw this draft and discussed parts of it with me. It bears the same title and contains, though not ordered in the best way, all the essential basic thoughts of the present writing, which is only their careful

¹Among the writings that have become known to me I mention E. Schröder's meritorious *Lehrbuch der Arithmetik und Algebra* (Leipzig 1873), in which one also finds a bibliography, and also the essays by Kronecker and by Helmholtz on the concept of number and on counting and measuring, in the collection of philosophical essays addressed to E. Zeller, Leipzig 1887. The appearance of these essays is the occasion which has moved me now to come forward also with my own conception, in some respects similar, but nevertheless essentially different through its foundation, which I have formed for myself for many years and without any influence from any side.

²Compare § 3 of my writing *Continuity and irrational numbers* (Braunschweig 1872).

³Dirichlet's *Vorlesungen über Zahlentheorie*, third edition, 1879, § 163, note on p. 470.

execution. As such main points I mention here the sharp distinction of the finite from the infinite (64), the concept of the number of things (161), the proof that the proof-method known as complete induction, or inference from n to $n + 1$, is indeed probative (59, 60, 80), and that definition by induction, or recursion, is also determinate and free of contradiction (126).

This writing can be understood by anyone who possesses what is called sound common sense; philosophical or mathematical schooling is not required in the least. But I know very well that many a person will scarcely recognize his numbers, which have accompanied him through his whole life as faithful and familiar friends, in the shadowy forms that I present to him. He will be put off and made impatient by the long series of simple inferences corresponding to the nature of our stepwise understanding, and by the sober analysis of the trains of thought on which the laws of numbers rest, when he is asked to follow proofs for truths that appear to him, according to his supposed inner intuition, evident and certain from the start. I, on the contrary, see precisely in the possibility of reducing such truths to others that are simpler – however long and apparently artificial the chain of inferences may be – a convincing proof that possession of them, or belief in them, is never given immediately by inner intuition, but is always acquired only by a more or less complete repetition of the individual inferences. I should like to compare this mental activity, difficult to follow because of the speed with which it is carried out, with the activity of a perfectly practiced reader when reading. This reading too always remains a more or less complete repetition of the individual steps that the beginner has to carry out in laborious spelling; but a very small part of them, and therefore a very small work or exertion of the mind, is already sufficient for the practiced reader to recognize the correct, true word, though only with very great probability. For it is well known that even the most practiced proofreader from time to time happens to let a misprint stand, that is, to read falsely, which would be impossible if the chain of thought belonging to spelling were completely repeated. Thus from our birth onward we are constantly, and in ever increasing measure, prompted to relate things to things and thereby to practice that capacity of the mind on which the creation of numbers also rests. Through this unceasing, though unintentional, practice already falling in our earliest years, and through the formation of judgments and chains of inference connected with it, we also acquire a store of properly arithmetical truths, to which later our first teachers appeal as to something simple, self-evident, given in inner intuition; and thus it comes about that some concepts that are in fact very composite, such as that of the number of things, falsely count as simple. In this sense, which I designate by the words modeled on a familiar saying, *man arithmetizes*, may the following pages, as an attempt to establish the science of numbers on a unified basis, receive a benevolent reception, and may they stimulate other mathematicians to reduce the long chains of inferences to a more modest and more agreeable measure.

In accordance with the purpose of this writing I restrict myself to the consideration of the sequence of the so-called natural numbers. In what way the gradual extension of the concept of number, the creation of zero, of negative, fractional, irrational, and complex numbers, is later to be brought about by reduction to earlier concepts, and indeed without any admixture of alien ideas, such as that of measurable magnitudes, which in my view can be raised to complete clarity only by the science of numbers – this I have shown at least in the example of irrational numbers in my earlier writing on continuity (1872). In quite similar fashion, as I also stated there (§ 3), the other extensions can be treated easily, and I reserve the right to devote a connected presentation to this subject. From precisely this

conception it appears self-evident and in no way new that every theorem of algebra and higher analysis, however remote, can be stated as a theorem about the natural numbers, an assertion that I repeatedly heard from Dirichlet's own mouth. But I do not see anything meritorious in actually undertaking this laborious rewriting and in wanting to use and recognize no numbers other than the natural numbers; that was also entirely far from Dirichlet's mind. On the contrary, the greatest and most fruitful advances in mathematics and other sciences have been made primarily through the creation and introduction of new concepts, after the frequent recurrence of composite phenomena, which can be mastered only laboriously by the old concepts, has pressed toward them. On this subject, in the summer of 1854, on the occasion of my habilitation as Privatdozent in Göttingen, I had to give a lecture before the philosophical faculty [LX], whose intention was also approved by Gauss; but this is not the place to go into that more closely.

Instead I use the opportunity to make a few remarks concerning my earlier writing, mentioned above, on *Continuity and irrational numbers*. The theory of irrational numbers presented there, conceived in the autumn of 1858, is founded on that phenomenon occurring in the domain of rational numbers (§ 4) to which I gave the name of a cut and which I first investigated exactly; it culminates in the proof of the continuity of the new domain of real numbers (§ 5. IV). It seems to me somewhat simpler, I would like to say calmer, than the two theories different from it and from one another which were set up by Messrs. Weierstrass and G. Cantor and likewise possess complete rigor. It was later taken up without essential change by Mr. U. Dini in the *Fondamenti per la teorica delle funzioni di variabili reali* (Pisa 1878); but the circumstance that my name is mentioned in the course of that presentation not in the description of the purely arithmetical phenomenon of the cut, but accidentally just where the existence of a measurable magnitude corresponding to the cut is at issue, could easily lead to the supposition that my theory rested on the consideration of such magnitudes. Nothing could be more incorrect. Rather, in § 3 of my writing I gave several reasons why I entirely reject the admixture of measurable magnitudes, and especially at the end, concerning their existence, I remarked that for a large part of the science of space the continuity of its formations is not even a necessary presupposition – quite apart from the fact that in works on geometry it is indeed mentioned incidentally by name, but never clearly explained, and therefore also not made accessible to proofs. To explain this more closely, I note for example the following. Choose arbitrarily three points A, B, C not lying on one straight line, with only the restriction that the ratios of their distances AB, AC, BC are algebraic⁴ numbers, and regard as present in space only those points M for which the ratios of AM, BM, CM to AB are likewise algebraic numbers. Then the space consisting of these points M is, as is easy to see, everywhere discontinuous; but despite the discontinuity and gappiness of this space, all constructions that occur in Euclid's *Elements*, so far as I can see, are executable in it exactly as in perfectly continuous space. The discontinuity of this space would therefore not be noticed at all, not be felt at all, in Euclid's science. But if someone tells me that we cannot think space otherwise than as continuous, I should like to doubt this and to point out how far advanced and refined a scientific cultivation is required merely to recognize clearly the nature of continuity and to understand that, besides rational magnitude-ratios, irrational ones are thinkable, and besides algebraic ones, transcendental ones. All the more beautiful it seems to me that man, without any idea of measurable magnitudes, and indeed through a finite system of simple steps of thought, can raise himself to the creation of the

⁴Dirichlet's *Vorlesungen über Zahlentheorie*, § 159 of the second, § 160 of the third edition.

pure, continuous realm of numbers; and only with this aid does it become possible, in my view, to form a clear idea of continuous space.

The same theory of irrational numbers, founded on the phenomenon of the cut, is also presented in J. Tannery's *Introduction à la théorie des fonctions d'une variable* (Paris 1886). If I understand one passage of the preface to this work correctly, the author conceived this theory independently, at a time when not only my writing, but also Dini's *Fondamenti* mentioned in that preface, was still unknown to him. This agreement seems to me a gratifying proof that my conception corresponds to the nature of the matter, as has also been acknowledged by other mathematicians, for example by Mr. M. Pasch in his *Einleitung in die Differential- und Integralrechnung* (Leipzig 1883). On the other hand, I cannot unreservedly agree with Mr. Tannery when he calls this theory the development of an idea due to Mr. J. Bertrand, contained in Bertrand's *Traité d'arithmétique*, and consisting in defining an irrational number by giving all rational numbers that are smaller than, and all those that are greater than, the number to be defined. Concerning this statement, which Mr. O. Stolz repeated – apparently without closer examination – in the preface to the second part of his *Vorlesungen über allgemeine Arithmetik* (Leipzig 1886), I allow myself only the following remark. That an irrational number is in fact to be regarded as completely determined by the specification just described was certainly, even before Mr. Bertrand, common property among all mathematicians who had occupied themselves with the concept of the irrational. Every calculator who approximately calculates an irrational root of an equation has precisely this mode of determination in mind; and when, as Mr. Bertrand does exclusively in his work (I have before me the eighth edition from the year 1885), one conceives an irrational number as the ratio of measurable magnitudes, then this mode of its determinateness is already stated most clearly in the famous definition that Euclid (*Elements* V. 5) gives for equality of ratios. This ancient conviction is certainly the source of my theory, as of Mr. Bertrand's and of many other more or less carried-out attempts to ground the introduction of irrational numbers into arithmetic. But even if one will agree completely with Mr. Tannery up to this point, a genuine examination must immediately observe that Mr. Bertrand's presentation, in which the phenomenon of the cut is not even mentioned in its logical purity, has no similarity at all with mine, inasmuch as it immediately takes refuge in the existence of a measurable magnitude, something I entirely reject for the reasons discussed above. And apart from this circumstance, this presentation also seems to me to exhibit, in the subsequent definitions and proofs founded on the assumption of that existence, several gaps so essential that I still hold justified, even with respect to this work, otherwise excellent in many ways and then unknown to me, the assertion made in my writing (§ 6) that the theorem $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$ had nowhere yet been rigorously proved.

Harzburg, 5 October 1887. R. Dedekind.

Preface to the second edition

Soon after its appearance the present writing found unfavorable as well as favorable assessments; indeed, serious errors were charged against it. I have not been able to convince myself of the correctness of these objections, and I now let the recently out-of-print writing, for whose public defense I lack the time, be reprinted without any change, adding only the following remarks to the first preface.

The property that I used as the definition (64) of an infinite system had already been emphasized before the appearance of my writing by G. Cantor ("Ein Beitrag zur Mannigfaltigkeitslehre," *Crelle's Journal*, vol. 84; 1878), indeed even by Bolzano (*Paradoxien des Unendlichen* § 20; 1851). But none of the named writers attempted to raise this property to the definition of the infinite and, on this foundation, to build the science of numbers rigorously logically; precisely in this consists the content of my laborious work, which in all essentials I had completed several years before the appearance of G. Cantor's paper and at a time when Bolzano's work was entirely unknown to me even by name. For those who have interest and understanding for the difficulties of such an investigation, I add the following. One can set up an entirely different definition of the finite and the infinite, which seems still simpler in that it does not even presuppose the concept of the similarity of a mapping (26), namely:

"A system S is called finite if it can be mapped into itself (36) in such a way that no proper part (6) of S is mapped into itself; in the opposite case S is called an infinite system."

Now let one once attempt to erect the structure on this new foundation! One will soon encounter great difficulties, and I believe I may assert that even the proof of complete agreement of this definition with the earlier one succeeds only when, and then easily, one may already regard the sequence of natural numbers as developed and also call upon the concluding consideration in (131); and yet none of all these things is mentioned either in the one definition or in the other! In this one recognizes how very great is the number of steps of thought required for such a transformation of a definition.

About a year after the publication of my writing I became acquainted with G. Frege's *Grundlagen der Arithmetik*, which had appeared already in 1884. However different the view of the essence of number set down in this work may be from mine, it nevertheless contains, especially from § 79 onward, very close points of contact with my writing, in particular with my explanation (44). To be sure, the agreement is not easy to recognize because of the different mode of expression; but the definiteness with which the author speaks about the inference from n to $n + 1$ (at the bottom of p. 93) clearly shows that here he stands on the same ground as I do.

Meanwhile (1890–1891) E. Schröder's *Vorlesungen über die Algebra der Logik* have appeared almost completely. It is impossible here to go more closely into the importance of this highly stimulating work, to which I offer my greatest recognition. Rather I should like only to excuse myself for nevertheless retaining my somewhat cumbersome notations (8) and (17), despite the remark made on p. 253 of the first part; they make no claim to be generally adopted, but modestly limit themselves to serving merely the purposes of this arithmetical writing, for which in my view they are better suited than summation and product signs.

Harzburg, 24 August 1893. R. Dedekind.

Preface to the third edition

When, about eight years ago, I was asked to replace the then already out-of-print second edition of this writing by a third, I hesitated to do so, because doubts had

meanwhile arisen about the security of important foundations of my conception. I do not fail to recognize the significance and partial justification of these doubts even today. But my trust in the inner harmony of our logic has not been shaken thereby. I believe that a rigorous investigation of the creative power of the mind – out of determinate elements to create a new determinate thing, their system, which is necessarily different from each of these elements – will certainly lead to making the foundations of my writing unobjectionable. Other work, however, prevents me from bringing such a difficult investigation to completion, and I therefore ask indulgence if the writing now nevertheless appears for the third time in unchanged form, something that can be justified only by the fact that interest in it, as the continuing demand shows, has not yet died out.

Braunschweig, 30 September 1911. R. Dedekind.

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§ 1. Systems of elements

1. In what follows I understand by a thing every object of our thought. In order to be able to speak conveniently of things, one denotes them by signs, for example by letters, and one permits oneself to speak briefly of the thing a , or even of a , where in truth one means the thing designated by a , not at all the letter a itself. A thing is completely determined by everything that can be asserted or thought of it. A thing a is the same as b (identical with b), and b the same as a , if everything that can be thought of a can also be thought of b , and if everything that holds of b can also be thought of a . That a and b are only signs or names for one and the same thing is indicated by the sign $a = b$ and likewise by $b = a$. If, moreover, $b = c$, so that c , like a , is likewise a sign for the thing designated by b , then $a = c$ also. If the agreement just described of the thing designated by a with the thing designated by b is not present, then these things a, b are called different: a is another thing than b , b another thing than a ; there is some property that belongs to the one and does not belong to the other.

2. It occurs very often that different things $a, b, c \dots$ are, for some reason, apprehended from a common point of view and brought together in the mind; one then says that they form a system S . The things $a, b, c \dots$ are called the elements of the

system S ; they are contained in S ; conversely, S consists of these elements. Such a system S (or a collection, a manifold, a totality) is likewise a thing (1) as an object of our thought; it is completely determined when, for every thing, it is determined whether it is an element of S or not.⁵ The system S is therefore the same as the system T , in signs $S = T$, if every element of S is also an element of T and every element of T is also an element of S . For uniformity of expression it is advantageous to allow also the special case in which a system S consists of a single element a (of one and only one element), that is, in which the thing a is an element of S , but every thing different from a is not an element of S . By contrast, for certain reasons we shall here exclude entirely the empty system, which contains no element at all, although for other investigations it can be convenient to invent such a system.

3. Definition. A system A is called a part of a system S if every element of A is also an element of S . Since this relation between a system A and a system S will come up again and again in what follows, we shall express it for brevity by the sign $A \subset S$. The converse sign $S \supset A$, by which the same fact could be designated, I shall, for the sake of clarity and simplicity, avoid entirely; but, for lack of a better word, I shall sometimes say that S is a whole of A , thereby intending to express that among the elements of S are also all elements of A . Since, moreover, every element s of a system S can itself be apprehended as a system, according to 2, we can apply the notation $s \subset S$ to this case as well.

4. Proposition. By 3, $A \subset A$.

5. Proposition. If $A \subset B$ and $B \subset A$, then $A = B$. The proof follows from 3 and 2.

6. Definition. A system A is called a proper part of S if A is part of S but is different from S . By 5, S is then not a part of A , that is, (3) there is in S an element which is not an element of A .

7. Proposition. If $A \subset B$ and $B \subset C$, which can also be denoted briefly by $A \subset B \subset C$, then $A \subset C$; and A is certainly a proper part of C if A is a proper part of B , or if B is a proper part of C . The proof follows from 3 and 6.

8. Definition. By the system composed from any systems $A, B, C \dots$, which is to be denoted by $\mathfrak{M}(A, B, C \dots)$, one understands that system whose elements are determined by the following prescription: a thing counts as an element of $\mathfrak{M}(A, B, C \dots)$ if and only if it is an element of one of the systems $A, B, C \dots$, that is, an element of A or B or $C \dots$. We also allow the case in which only a single system A is present; then, evidently, $\mathfrak{M}(A) = A$. We further note that the system $\mathfrak{M}(A, B, C \dots)$ composed from $A, B, C \dots$ is to be carefully distinguished from that system whose elements are the systems $A, B, C \dots$ themselves.

9. Proposition. The systems $A, B, C \dots$ are parts of $\mathfrak{M}(A, B, C \dots)$. The proof follows from 8 and 3.

10. Proposition. If $A, B, C \dots$ are parts of a system S , then $\mathfrak{M}(A, B, C \dots) \subset S$. The proof follows from 8 and 3.

11. Proposition. If P is part of one of the systems $A, B, C \dots$, then $P \subset \mathfrak{M}(A, B, C \dots)$. The proof follows from 9 and 7.

⁵In what way this determinateness comes about, and whether we know a way of deciding it, is wholly indifferent for everything that follows; the general laws to be developed do not depend on this at all, and hold under all circumstances. I mention this expressly because Herr Kronecker has recently (in volume 99 of the *Journal für Mathematik*, pp. 334–336) wished to impose certain restrictions on free formation of concepts in mathematics, restrictions which I do not recognize as justified; to go into this more closely, however, will appear necessary only when that distinguished mathematician has published his reasons for the necessity, or even merely the expediency, of these restrictions.

12. Proposition. If each of the systems $P, Q \dots$ is part of one of the systems $A, B, C \dots$, then $\mathfrak{M}(P, Q \dots) \subset \mathfrak{M}(A, B, C \dots)$. The proof follows from 11 and 10.

13. Proposition. If A is composed from any of the systems $P, Q \dots$, then $A \subset \mathfrak{M}(P, Q \dots)$. **Proof.** For every element of A is, by 8, an element of one of the systems $P, Q \dots$, and hence, again by 8, also an element of $\mathfrak{M}(P, Q \dots)$, from which the proposition follows by 3.

14. Proposition. If each of the systems $A, B, C \dots$ is composed from any of the systems $P, Q \dots$, then $\mathfrak{M}(A, B, C \dots) \subset \mathfrak{M}(P, Q \dots)$. The proof follows from 13 and 10.

15. Proposition. If each of the systems $P, Q \dots$ is part of one of the systems $A, B, C \dots$, and each of the latter is composed from some of the former, then $\mathfrak{M}(P, Q \dots) = \mathfrak{M}(A, B, C \dots)$. The proof follows from 12, 14, and 5.

16. Proposition. If $A = \mathfrak{M}(P, Q)$ and $B = \mathfrak{M}(Q, R)$, then $\mathfrak{M}(A, R) = \mathfrak{M}(P, B)$. **Proof.** For by the preceding Proposition 15, both $\mathfrak{M}(A, R)$ and $\mathfrak{M}(P, B)$ are equal to $\mathfrak{M}(P, Q, R)$.

17. Definition. A thing g is called a common element of the systems $A, B, C \dots$ if it is contained in each of these systems (thus in A and in B and in $C \dots$). Likewise, a system T is called a common part of $A, B, C \dots$ if T is part of each of these systems; and by the commonality of the systems $A, B, C \dots$ we understand the completely determined system $\mathfrak{G}(A, B, C \dots)$, which consists of all common elements g of $A, B, C \dots$ and consequently is itself also a common part of those systems. We again allow the case in which only one single system A is present; then $\mathfrak{G}(A) = A$ is to be set. But the case can also occur in which the systems $A, B, C \dots$ have no common element at all, and therefore possess no common part, no commonality; they are then called systems without a common part, and the sign $\mathfrak{G}(A, B, C \dots)$ is meaningless (compare the end of 2). We shall, however, almost always leave it to the reader, in propositions about commonalities, to supply mentally the condition of their existence and to find the right interpretation of these propositions also for the case of non-existence.

18. Proposition. Every common part of $A, B, C \dots$ is part of $\mathfrak{G}(A, B, C \dots)$. The proof follows from 17.

19. Proposition. Every part of $\mathfrak{G}(A, B, C \dots)$ is a common part of $A, B, C \dots$. The proof follows from 17 and 7.

20. Proposition. If each of the systems $A, B, C \dots$ is a whole (3) of one of the systems $P, Q \dots$, then $\mathfrak{G}(P, Q \dots) \subset \mathfrak{G}(A, B, C \dots)$. **Proof.** For every element of $\mathfrak{G}(P, Q \dots)$ is a common element of $P, Q \dots$, hence also a common element of $A, B, C \dots$, as was to be proved.

§ 2. Mapping of a system

21. Definition.⁶ By a mapping φ of a system S one understands a law according to which, to each determinate element s of S , there belongs a determinate thing, which is called the image of s and is denoted by $\varphi(s)$. We also say that $\varphi(s)$ corresponds to the element s , that $\varphi(s)$ arises or is generated from s by the mapping φ , and that s passes over into $\varphi(s)$ under the mapping φ . If now T is any part of S , then the mapping φ of S contains at the same time a determinate mapping of T , which, for the sake of simplicity, may well be denoted by the same sign φ and consists in this: to every element t of the system T there corresponds the same image $\varphi(t)$ that t possesses as an element of S . At the same time, the system consisting of all

⁶Compare Dirichlet's *Vorlesungen über Zahlentheorie*, third edition, 1879, § 163.

images $\varphi(t)$ is to be called the image of T and denoted by $\varphi(T)$; this also explains the meaning of $\varphi(S)$. As an example of a mapping of a system one may already regard the assignment of determinate signs or names to its elements. The simplest mapping of a system is that by which each of its elements passes over into itself; it is to be called the identical mapping of the system. For convenience, in the following Propositions 22, 23, 24, which refer to an arbitrary mapping φ of an arbitrary system S , we shall denote the images of elements s and of parts T respectively by s' and T' ; in addition, we stipulate that small and capital Latin letters without an accent are always to mean elements and parts of this system S .

22. Proposition.⁷ If $A \subset B$, then $A' \subset B'$. **Proof.** For every element of A' is the image of an element contained in A , hence also in B , and is consequently an element of B' , as was to be proved.

23. Proposition. The image of $\mathfrak{M}(A, B, C \dots)$ is $\mathfrak{M}(A', B', C' \dots)$. **Proof.** Denote by M the system $\mathfrak{M}(A, B, C \dots)$, which by 10 is likewise part of S . Then every element of its image M' is the image m' of an element m of M ; since m , by 8, is also an element of one of the systems $A, B, C \dots$, m' is consequently an element of one of the systems $A', B', C' \dots$, and hence by 8 also an element of $\mathfrak{M}(A', B', C' \dots)$. Thus, by 3, $M' \subset \mathfrak{M}(A', B', C' \dots)$. Conversely, since $A, B, C \dots$ are, by 9, parts of M , $A', B', C' \dots$ are by 22 parts of M' ; hence by 10 also $\mathfrak{M}(A', B', C' \dots) \subset M'$. Combining this with the preceding result gives, by 5, the proposition to be proved: $M' = \mathfrak{M}(A', B', C' \dots)$.

24. Proposition.⁸ The image of every common part of $A, B, C \dots$, and therefore also that of the commonality $\mathfrak{G}(A, B, C \dots)$, is part of $\mathfrak{G}(A', B', C' \dots)$. **Proof.** For by 22 it is a common part of $A', B', C' \dots$, from which the proposition follows by 18.

25. Definition and proposition. If φ is a mapping of a system S , and ψ a mapping of the image $S' = \varphi(S)$, then there always arises from φ and ψ a composite⁹ mapping θ of S , consisting in this: to every element s of S there corresponds the image

$$\theta(s) = \psi(s') = \psi(\varphi(s)),$$

where again $\varphi(s) = s'$ is set. This mapping θ can be denoted briefly by the symbol $\psi \cdot \varphi$ or $\psi\varphi$, and the image $\theta(s)$ by $\psi\varphi(s)$; attention must be paid to the order of the signs φ, ψ , because the sign $\varphi\psi$ is in general meaningless and has sense only when $\psi(S') \subset S$. If χ denotes a mapping of the system $\psi(S')$, and η the mapping $\chi\psi$ of the system S' composed from ψ and χ , then

$$\chi\theta(s) = \chi\psi(s') = \eta(s') = \eta\varphi(s),$$

that is, for every element s of S the composite mappings $\chi\theta = \chi\psi\varphi$ and $\eta\varphi$ agree with one another; hence $\chi\theta = \eta\varphi$. According to the meaning of θ and η , this proposition can fittingly be expressed by

$$\chi \cdot \psi\varphi = \chi\psi \cdot \varphi,$$

and this mapping composed from φ, ψ, χ can be denoted briefly by $\chi\psi\varphi$.

§ 3. Similarity of a mapping. Similar systems

26. Definition. A mapping φ of a system S is called similar (or distinct) if different elements a, b of the system S always correspond to different images $a' = \varphi(a)$,

⁷ Compare Proposition 27.

⁸ Compare Proposition 29.

⁹ A confusion of this composition of mappings with the composition of systems of elements (8) is presumably not to be feared.

$b' = \varphi(b)$. Since in this case, conversely, $s' = t'$ always implies $s = t$, every element of the system $S' = \varphi(S)$ is the image s' of one single, completely determined element s of the system S ; hence one can place opposite to the mapping φ of S an inverse mapping of the system S' , to be denoted for instance by $\bar{\varphi}$, which consists in this: to each element s' of S' there corresponds the image $\bar{\varphi}(s') = s$, and this mapping is evidently likewise similar. It is clear that $\bar{\varphi}(S') = S$, that φ is the inverse mapping belonging to $\bar{\varphi}$, and that the mapping $\bar{\varphi}\varphi$ composed from φ and $\bar{\varphi}$ according to 25 is the identical mapping of S (21). At the same time, the following supplements to § 2 result, with the notation used there retained.

27. Proposition.¹⁰ If $A' \subset B'$, then $A \subset B$. **Proof.** For if a is an element of A , then a' is an element of A' , hence also of B' , and therefore $a' = b'$, where b is an element of B ; but since $a' = b'$ always implies $a = b$, every element a of A is also an element of B , as was to be proved.

28. Proposition. If $A' = B'$, then $A = B$. The proof follows from 27, 4, and 5.

29. Proposition.¹¹ If $G = \mathfrak{S}(A, B, C \dots)$, then $G' = \mathfrak{S}(A', B', C' \dots)$. **Proof.** Every element of $\mathfrak{S}(A', B', C' \dots)$ is in any case contained in S' , and so is the image g' of an element g contained in S . But since g' is a common element of $A', B', C' \dots$, g must by 27 be a common element of $A, B, C \dots$, and therefore also an element of G . Hence every element of $\mathfrak{S}(A', B', C' \dots)$ is the image of an element g of G , and so is an element of G' ; that is, $\mathfrak{S}(A', B', C' \dots) \subset G'$. Taking account of 24 and 5, our proposition follows.

30. Proposition. The identical mapping of a system is always a similar mapping.

31. Proposition. If φ is a similar mapping of S , and ψ a similar mapping of $\varphi(S)$, then the mapping $\psi\varphi$ of S composed from φ and ψ is likewise similar, and the corresponding inverse mapping $\bar{\psi\varphi}$ is $= \bar{\varphi}\bar{\psi}$. **Proof.** For to different elements a, b of S there correspond different images $a' = \varphi(a)$, $b' = \varphi(b)$, and to these in turn different images $\psi(a') = \psi\varphi(a)$, $\psi(b') = \psi\varphi(b)$; hence $\psi\varphi$ is a similar mapping. Moreover, every element $\psi\varphi(s) = \psi(s')$ of the system $\psi\varphi(S)$ passes under $\bar{\psi}$ into $s' = \varphi(s)$, and this under $\bar{\varphi}$ into s ; thus $\psi\varphi(s)$ passes under $\bar{\varphi}\bar{\psi}$ into s , as was to be proved.

32. Definition. The systems R, S are called similar if there is a similar mapping φ of S such that $\varphi(S) = R$, and hence also $\bar{\varphi}(R) = S$. Evidently, by 30, every system is similar to itself.

33. Proposition. If R, S are similar systems, then every system Q similar to R is also similar to S . **Proof.** For if φ, ψ are similar mappings of S, R such that $\varphi(S) = R$ and $\psi(R) = Q$, then by 31 $\psi\varphi$ is a similar mapping of S such that $\psi\varphi(S) = Q$, as was to be proved.

34. Definition. One can therefore divide all systems into classes, by putting into a determinate class all and only those systems $Q, R, S \dots$ that are similar to a determinate system R , the representative of the class. By the preceding Proposition 33, the class does not change if any other system S belonging to it is chosen as representative.

35. Proposition. If R, S are similar systems, then every part of S is similar to a part of R , and every proper part of S is similar to a proper part of R . **Proof.** For if φ is a similar mapping of S , $\varphi(S) = R$, and $T \subset S$, then by 22 the system $\varphi(T) \subset R$, which is similar to T , is a part of R . If, furthermore, T is a proper part of S , and s is an element of S not contained in T , then by 27 the element $\varphi(s)$ contained in R cannot be contained in $\varphi(T)$; hence $\varphi(T)$ is a proper part of R , as was to be proved.

¹⁰Compare Proposition 22.

¹¹Compare Proposition 24.

§ 4. Mapping of a system into itself

36. Definition. If φ is a similar or non-similar mapping of a system S , and $\varphi(S)$ is part of a system Z , then we call φ a mapping of S into Z , and we say that S is mapped by φ into Z . We therefore call φ a mapping of the system S into itself if $\varphi(S) \subset S$, and in this section we shall investigate the general laws of such a mapping φ . Here we use the same notations as in § 2, again putting $\varphi(s) = s'$, $\varphi(T) = T'$. These images s', T' are now, by 22 and 7, again themselves elements or parts of S , as are all things denoted by Latin letters.

37. Definition. K is called a *chain* if $K' \subset K$. We expressly note that this name is not given to the part K of the system S in itself, but only in relation to the determinate mapping φ ; with respect to another mapping of the system S into itself, K may very well not be a chain.

38. Proposition. S is a chain.

39. Proposition. The image K' of a chain K is a chain. **Proof.** For from $K' \subset K$ it follows, by 22, that $(K')' \subset K'$, as was to be proved.

40. Proposition. If A is part of a chain K , then $A' \subset K$ also. **Proof.** For from $A \subset K$ it follows (by 22) that $A' \subset K'$, and since (by 37) $K' \subset K$, it follows (by 7) that $A' \subset K$, as was to be proved.

41. Proposition. If the image A' is part of a chain L , then there is a chain K satisfying the conditions $A \subset K$, $K' \subset L$; indeed, $\mathfrak{M}(A, L)$ is such a chain K . **Proof.** If one actually sets $K = \mathfrak{M}(A, L)$, then by 9 the first condition $A \subset K$ is fulfilled. Since, by 23, $K' = \mathfrak{M}(A', L')$, and by hypothesis $A' \subset L$, $L' \subset L$, by 10 the other condition $K' \subset L$ is also fulfilled; and from this, since (by 9) $L \subset K$, it also follows that $K' \subset K$, that is, K is a chain, as was to be proved.

42. Proposition. A system M composed entirely of chains $A, B, C \dots$ is a chain. **Proof.** Since (by 23) $M' = \mathfrak{M}(A', B', C' \dots)$, and by hypothesis $A' \subset A$, $B' \subset B$, $C' \subset C \dots$, it follows (by 12) that $M' \subset M$, as was to be proved.

43. Proposition. The commonality G of chains $A, B, C \dots$ is a chain. **Proof.** Since G is, by 17, a common part of $A, B, C \dots$, G' is by 22 a common part of $A', B', C' \dots$; and since by hypothesis $A' \subset A$, $B' \subset B$, $C' \subset C \dots$, G' is (by 7) also a common part of $A, B, C \dots$ and consequently by 18 also part of G , as was to be proved.

44. Definition. If A is any part of S , then by A_0 we shall denote the commonality of all those chains (for example S) of which A is part; this commonality A_0 exists (compare 17), since A itself is a common part of all these chains. Since, furthermore, A_0 is a chain by 43, we shall call A_0 the chain of the system A , or briefly the chain of A . This definition too refers entirely to the determinate underlying mapping φ of the system S into itself; and if later clarity requires it, then instead of A_0 we shall prefer the notation $\varphi_0(A)$, and likewise we shall denote by $\omega_0(A)$ the chain of A corresponding to another mapping ω . The following propositions now hold for this very important concept.

45. Proposition. $A \subset A_0$. **Proof.** For A is a common part of all those chains whose commonality is A_0 , from which the proposition follows by 18.

46. Proposition. $(A_0)' \subset A_0$. **Proof.** For by 44, A_0 is a chain (37).

47. Proposition. If A is part of a chain K , then $A_0 \subset K$. **Proof.** For A_0 is the commonality, and hence also a common part, of all chains K of which A is part.

48. Remark. One easily convinces oneself that the concept of the chain A_0 defined in 44 is completely characterized by the preceding propositions 45, 46, and 47.

49. Proposition. $A' \subset (A_0)'$. The proof follows from 45 and 22.

50. Proposition. $A' \subset A_0$. The proof follows from 49, 46, and 7.

51. Proposition. If A is a chain, then $A_0 = A$. **Proof.** Since A is part of the chain A , by 47 also $A_0 \subset A$, from which, by 45 and 5, the proposition follows.

52. Proposition. If $B \subset A$, then $B \subset A_0$. The proof follows from 45 and 7.

53. Proposition. If $B \subset A_0$, then $B_0 \subset A_0$, and conversely. **Proof.** Since A_0 is a chain, from $B \subset A_0$ it follows by 47 that $B_0 \subset A_0$; conversely, if $B_0 \subset A_0$, then by 7 also $B \subset A_0$, because (by 45) $B \subset B_0$.

54. Proposition. If $B \subset A$, then $B_0 \subset A_0$. The proof follows from 52 and 53.

55. Proposition. If $B \subset A_0$, then also $B' \subset A_0$. **Proof.** For by 53, $B_0 \subset A_0$, and since (by 50) $B' \subset B_0$, the proposition to be proved follows from 7. The same result, as is easy to see, follows also from 22, 46, 7, or also from 40.

56. Proposition. If $B \subset A_0$, then $(B_0)' \subset (A_0)'$. The proof follows from 53 and 22.

57. Proposition and definition. One has

$$(A_0)' = (A')_0,$$

that is, the image of the chain of A is at the same time the chain of the image of A . Thus one can denote this system briefly by A'_0 and call it, as one chooses, the chain-image or the image-chain of A . In the clearer notation indicated in 44, the proposition would be expressed by

$$\varphi(\varphi_0(A)) = \varphi_0(\varphi(A)).$$

Proof. For abbreviation set $(A')_0 = L$. Then L is a chain (44), and by 45 $A' \subset L$; hence by 41 there is a chain K satisfying the conditions $A \subset K$, $K' \subset L$. From this it follows by 47 that $A_0 \subset K$, hence $(A_0)' \subset K'$, and consequently by 7 also

$$(A_0)' \subset (A')_0.$$

Since, moreover, by 49, $A' \subset (A_0)'$, and $(A_0)'$ is a chain by 44 and 39, by 47 one also has

$$(A')_0 \subset (A_0)'.$$

Combining this with the preceding result gives the proposition to be proved (5).

58. Proposition. One has

$$A_0 = \mathfrak{M}(A, A'_0),$$

that is, the chain of A is composed of A and the image-chain of A .

Proof. Again put, for abbreviation,

$$L = A'_0 = (A_0)' = (A')_0 \quad \text{and} \quad K = \mathfrak{M}(A, L).$$

Then (by 45) $A' \subset L$, and since L is a chain, the same holds for K by 41; furthermore, since $A \subset K$ (9), by 47 also

$$A_0 \subset K.$$

Conversely, since (by 45) $A \subset A_0$ and by 46 also $L \subset A_0$, by 10 also

$$K \subset A_0.$$

Combining this with the preceding result gives the proposition $A_0 = K$ (5).

59. Proposition of complete induction. In order to prove that the chain A_0 is part of any system Σ — whether or not the latter is part of S — it is enough to show:

ρ . $A \subset \Sigma$, and

σ . the image of every common element of A_0 and Σ is likewise an element of Σ .

Proof. For if ρ is true, then by 45 the commonality $G = \mathfrak{G}(A_0, \Sigma)$ certainly exists, and indeed (by 18) $A \subset G$; since moreover by 17

$$G \subset A_0,$$

G is also part of our system S , which is mapped into itself by φ , and at the same time by 55 also $G' \subset A_0$. If now σ is also true, that is, if $G' \subset \Sigma$, then G' , as a common part of the systems A_0, Σ , must by 18 be part of their commonality G ; that is, G is a chain (37). Since, as already noted above, $A \subset G$, by 47 also

$$A_0 \subset G.$$

Together with the preceding result this gives $G = A_0$, hence by 17 also $A_0 \subset \Sigma$, as was to be proved.

60. The preceding proposition forms, as will later be seen, the scientific foundation for the method of proof known under the name of complete induction (the inference from n to $n + 1$), and it can also be expressed in the following way: to prove that all elements of the chain A_0 possess a certain property $\mathcal{E}$ (or that a proposition S , in which an indeterminate thing n is spoken of, really holds for all elements n of the chain A_0), it is enough to show:

ρ . all elements a of the system A possess the property $\mathcal{E}$ (or S holds for all a), and

σ . to the image n' of every such element n of A_0 that possesses $\mathcal{E}$, the same property $\mathcal{E}$ belongs (or the proposition S , as soon as it holds for an element n of A_0 , must certainly also hold for its image n').

Indeed, if one denotes by Σ the system of all things that possess the property $\mathcal{E}$ (or for which the proposition S holds), then the complete agreement of the present wording of the proposition with that used in 59 is immediately evident.

61. Proposition. The chain of $\mathfrak{M}(A, B, C \dots)$ is $\mathfrak{M}(A_0, B_0, C_0 \dots)$. **Proof.** Denote the former system by M and the latter by K . Then K is a chain by 42. Since each of the systems $A, B, C \dots$ is, by 45, part of one of the systems $A_0, B_0, C_0 \dots$, it follows (by 12) that $M \subset K$, hence by 47 also

$$M_0 \subset K.$$

Conversely, since by 9 each of the systems $A, B, C \dots$ is part of M , and hence by 45 and 7 also part of the chain M_0 , by 47 each of the systems $A_0, B_0, C_0 \dots$ must also be part of M_0 , and hence by 10

$$K \subset M_0.$$

Combining this with the preceding result gives the proposition $M_0 = K$ (5).

62. Proposition. The chain of $\mathfrak{G}(A, B, C \dots)$ is part of $\mathfrak{G}(A_0, B_0, C_0 \dots)$. **Proof.** Denote the former system by G and the latter by K . Then K is a chain by 43. Since each of the systems $A_0, B_0, C_0 \dots$ is, by 45, a whole of one of the systems $A, B, C \dots$, it follows (by 20) that $G \subset K$; hence by 47 the proposition to be proved follows: $G_0 \subset K$.

63. Proposition. If $K' \subset L \subset K$, so that K is a chain, then L is also a chain. If L is a proper part of K , and U is the system of all those elements of K that are not contained in L , if furthermore the chain U_0 is a proper part of K , and V is the system of all those elements of K that are not contained in U_0 , then

$$K = \mathfrak{M}(U_0, V) \quad \text{and} \quad L = \mathfrak{M}(U'_0, V).$$

Finally, if $L = K'$, then $V \subset V'$.

The proof of this proposition, of which we shall make no use (as also of the two preceding ones), may be left to the reader.

§ 5. The finite and the infinite

64. Definition.¹² A system S is called infinite if it is similar to a proper part of itself (32); in the contrary case S is called a finite system.

65. Proposition. Every system consisting of a single element is finite. **Proof.** For such a system has no proper part at all (2, 6).

66. Proposition. Infinite systems exist. **Proof.**¹³ My world of thoughts, that is, the totality S of all things that can be objects of my thinking, is infinite. For if s denotes an element of S , then the thought s' that s can be an object of my thinking is itself an element of S . If one regards this as the image $\varphi(s)$ of the element s , then the mapping φ of S thereby determined has the property that the image S' is part of S ; and indeed S' is a proper part of S , because in S there are elements (for example my own self) that are different from every such thought s' and are therefore not contained in S' . Finally, it is evident that if a, b are different elements of S , then their images a', b' are also different; hence the mapping φ is a distinct (similar) one (26). Thus S is infinite, as was to be proved.

67. Proposition. If R, S are similar systems, then R is finite or infinite according as S is finite or infinite. **Proof.** If S is infinite, hence similar to a proper part S' of itself, then, if R and S are similar, S' must by 33 be similar to R and by 35 at the same time similar to a proper part of R , which consequently by 33 is itself similar to R ; hence R is infinite, as was to be proved.

68. Proposition. Every system S that possesses an infinite part T is itself infinite; or, in other words, every part of a finite system is finite. **Proof.** If T is infinite, then there is a similar mapping ψ of T such that $\psi(T)$ becomes a proper part of T . If T is part of S , one can extend this mapping ψ to a mapping φ of S by putting, when s denotes any element of S , $\varphi(s) = \psi(s)$ or $\varphi(s) = s$, according as s is an element of T or not. This mapping φ is similar. Namely, if a, b denote different elements of S , then, if both are contained in T , the image $\varphi(a) = \psi(a)$ is different from the image $\varphi(b) = \psi(b)$, because ψ is a similar mapping; if furthermore a is contained in T and b is not contained in T , then $\varphi(a) = \psi(a)$ is different from $\varphi(b) = b$, because $\psi(a)$ is contained in T ; and finally, if neither a nor b is contained in T , then also $\varphi(a) = a$ is different from $\varphi(b) = b$, as was to be shown. Since moreover $\psi(T)$ is part of T ,

¹²If one does not wish to use the concept of similar systems (32), then one must say: S is called infinite if there is a proper part of S (6) in which S can be distinctly (similarly) mapped (26, 36). In this form I communicated the definition of the infinite, which forms the core of my whole investigation, to Mr. G. Cantor in September 1882 and already several years earlier also to Messrs. Schwarz and Weber. All other attempts known to me to distinguish the infinite from the finite seem to me so little successful that I believe I may refrain from criticizing them.

¹³A similar consideration is found in § 13 of Bolzano's *Paradoxien des Unendlichen* (Leipzig 1851).

hence by 7 also part of S , it is evident that $\varphi(S) \subset S$ as well. Finally, since $\psi(T)$ is a proper part of T , there is in T , hence also in S , an element t that is not contained in $\psi(T) = \varphi(T)$; since the image $\varphi(s)$ of every element s not contained in T is s itself, and hence is also different from t , t cannot be contained in $\varphi(S)$ at all. Thus $\varphi(S)$ is a proper part of S , and consequently S is infinite, as was to be proved.

69. Proposition. Every system similar to a part of a finite system is itself finite. The proof follows from 67 and 68.

70. Proposition. If a is an element of S , and the aggregate T of all elements of S different from a is finite, then S is finite too. **Proof.** We have to show (by 64) that if φ denotes any similar mapping of S into itself, then the image $\varphi(S)$, or S' , is never a proper part of S , but is always $= S$. Evidently $S = \mathfrak{M}(a, T)$, and consequently, by 23, if the images are again denoted by accents, $S' = \mathfrak{M}(a', T')$; and because the mapping φ is similar, a' is not contained in T' (26). Since, furthermore, by hypothesis $S' \subset S$, a' and likewise every element of T' must either be $= a$ or an element of T . If therefore — this is the case we shall first treat — a is not contained in T' , then $T' \subset T$ and hence $T' = T$, because φ is a similar mapping and T is a finite system; and since a' , as noted, is not contained in T' , that is, not in T , we must have $a' = a$, and hence in this case indeed $S' = S$, as asserted. In the opposite case, when a is contained in T' and consequently is the image b' of an element b contained in T , let U denote the aggregate of all those elements u of T that are different from b ; then $T = \mathfrak{M}(b, U)$ and (by 15) $S = \mathfrak{M}(a, b, U)$, hence $S' = \mathfrak{M}(a', a, U')$. We now determine a new mapping ψ of T by putting $\psi(b) = a'$ and, in general, $\psi(u) = u'$, whereby (by 23) $\psi(T) = \mathfrak{M}(a', U')$. Evidently ψ is a similar mapping, because φ was such a mapping and because a is not contained in U , hence also a' is not contained in U' . Since, furthermore, a and every element u are different from b , by the similarity of φ both a' and every element u' must be different from a , and hence contained in T ; thus $\psi(T) \subset T$, and since T is finite, it must be that $\psi(T) = T$, hence $\mathfrak{M}(a', U') = T$. But from this it follows (by 15) that

$$\mathfrak{M}(a', a, U') = \mathfrak{M}(a, T),$$

that is, by the above, $S' = S$. Thus the required proof is carried out in this case as well.

§ 6. Simply infinite systems. The sequence of natural numbers

71. Definition. A system N is called simply infinite if there is a similar mapping φ of N into itself such that N appears as the chain (44) of an element that is not contained in $\varphi(N)$. We call this element, which in what follows we shall denote by the symbol 1, the base-element of N , and we say at the same time that the simply infinite system N is ordered by this mapping φ . Retaining the earlier convenient notations for the images and chains (§ 4), the essence of a simply infinite system N therefore consists in the existence of a mapping φ of N and of an element 1 satisfying the following conditions $\alpha, \beta, \gamma, \delta$:

$$\alpha. \quad N' \subset N,$$

$$\beta. \quad N = 1_0,$$

$$\gamma. \quad \text{the element 1 is not contained in } N',$$

$$\delta. \quad \text{the mapping } \varphi \text{ is similar.}$$

It is evident from α, γ, δ that every simply infinite system N is indeed an infinite system (64), because it is similar to a proper part N' of itself.

72. Proposition. In every infinite system S a simply infinite system N is contained as a part. **Proof.** By 64 there is a similar mapping φ of S such that $\varphi(S)$, or S' , becomes a proper part of S ; hence there is an element 1 in S that is not contained in S' . The chain $N = 1_0$ corresponding to this mapping φ of the system S into itself (44) is a simply infinite system ordered by φ ; for the characteristic conditions $\alpha, \beta, \gamma, \delta$ in 71 are evidently all fulfilled.

73. Definition. If, in considering a simply infinite system N ordered by a mapping φ , one entirely disregards the particular nature of the elements, retains only their distinguishability, and apprehends only the relations in which they are placed to one another by the ordering mapping φ , then these elements are called natural numbers or ordinal numbers, or simply numbers, and the base-element 1 is called the base-number of the number-sequence N . In view of this liberation of the elements from every other content (abstraction), one may rightly call the numbers a free creation of the human mind. The relations or laws that are derived solely from the conditions $\alpha, \beta, \gamma, \delta$ in 71, and that are therefore always the same in all ordered simply infinite systems, whatever names may happen to be given to the individual elements (compare 134), form the immediate subject of the science of numbers, or arithmetic. From the general concepts and propositions of § 4 on the mapping of a system into itself, we first take directly the following fundamental principles, where $a, b \dots m, n \dots$ always denote elements of N , hence numbers, $A, B, C \dots$ parts of N , and $a', b' \dots m', n' \dots A', B', C' \dots$ the corresponding images that are generated by the ordering mapping φ and are always again elements or parts of N ; the image n' of a number n is also called the number following n .

74. Proposition. By 45 every number n is contained in its chain n_0 , and by 53 the condition $n \subset m_0$ is equivalent to $n_0 \subset m_0$.

75. Proposition. By 57,

$$n'_0 = (n_0)' = (n')_0.$$

76. Proposition. By 46,

$$n'_0 \subset n_0.$$

77. Proposition. By 58,

$$n_0 = \mathfrak{M}(n, n'_0).$$

78. Proposition. $N = \mathfrak{M}(1, N')$; hence every number different from the base-number 1 is an element of N' , that is, the image of a number. The proof follows from 77 and 71.

79. Proposition. N is the only number-chain in which the base-number 1 is contained. **Proof.** For if 1 is an element of a number-chain K , then by 47 the corresponding chain $N \subset K$; hence $N = K$, since of course $K \subset N$.

80. Proposition of complete induction (inference from n to n'). To prove that a proposition holds for all numbers n of a chain m_0 , it is enough to show:

ρ . it holds for $n = m$, and

σ . from the validity of the proposition for a number n of the chain m_0 , its validity for the following number n' always follows.

This follows immediately from the more general Proposition 59 or 60. Most frequently the case will occur in which $m = 1$, and hence m_0 is the full number-sequence N .

§ 7. Greater and smaller numbers

81. Proposition. Every number n is different from the number n' following it. **Proof by complete induction (80).** For

$\rho.$

the proposition is true for the number $n = 1$, because it is not contained in N' (71), whereas the following number $1'$, as the image of the number 1 contained in N , is an element of N' .

$\sigma.$

If the proposition is true for a number n , and if the following number $n' = p$ is set, then n is different from p ; from this, by 26, because of the similarity (71) of the ordering mapping φ , it follows that p is different from p' . Thus the proposition also holds for the number p following n , as was to be proved.

82. Proposition. In the image-chain n'_0 of a number n , its image n' is indeed contained (by 74, 75), but the number n itself is not. **Proof by complete induction (80).** For

$\rho.$

the proposition is true for $n = 1$, because $1'_0 = N'$, and because by 71 the base-number 1 is not contained in N' .

$\sigma.$

If the proposition is true for a number n , and if again $n' = p$, then n is not contained in p_0 , hence is different from every number q contained in p_0 . From this, because of the similarity of φ , it follows that n' , hence p , is different from every number q' contained in p'_0 , and therefore is not contained in p'_0 . Thus the proposition also holds for the number p following n , as was to be proved.

83. Proposition. The image-chain n'_0 is a proper part of the chain n_0 . The proof follows from 76, 74, and 82.

84. Proposition. From $m_0 = n_0$ it follows that $m = n$. **Proof.** Since (by 74) m is contained in m_0 , and

$$m_0 = n_0 = \mathfrak{M}(n, n'_0)$$

(77), if the proposition were false, and hence m were different from n , then m would have to be contained in the chain n'_0 , and consequently by 74 also $m_0 \subset n'_0$, that is, $n_0 \subset n'_0$; since this contradicts Proposition 83, our proposition is proved.

85. Proposition. If the number n is not contained in the number-chain K , then $K \subset n'_0$. **Proof by complete induction (80).** For

$\rho.$

the proposition is true for $n = 1$ by 78.

$\sigma.$

If the proposition is true for a number n , then it also holds for the following number $p = n'$; for if p is not contained in the number-chain K , then by 40 also n cannot be

contained in K , and consequently by our assumption $K \subset n'_0$. Since now (by 77) $n'_0 = p_0 = \mathfrak{M}(p, p'_0)$, hence $K \subset \mathfrak{M}(p, p'_0)$, and p is not contained in K , it must be that $K \subset p'_0$, as was to be proved.

86. Proposition. If the number n is not contained in the number-chain K , but its image n' is contained in it, then $K = n'_0$. **Proof.** Since n is not contained in K , by 85 $K \subset n'_0$; and since $n' \in K$, by 47 also $n'_0 \subset K$, hence $K = n'_0$, as was to be proved.

87. Proposition. In every number-chain K there is one, and (by 84) only one, number k whose chain $k_0 = K$. **Proof.** If the base-number 1 is contained in K , then (by 79) $K = N = 1_0$. In the contrary case, let Z be the system of all numbers not contained in K ; since the base-number 1 is contained in Z , but Z is only a proper part of the number-sequence N , by 79 Z cannot be a chain, that is, Z' cannot be part of Z ; hence in Z there is a number n whose image n' is not in Z , and therefore is certainly contained in K . Since, moreover, n is in Z , hence not contained in K , by 86 $K = n'_0$, and thus $k = n'$, as was to be proved.

88. Proposition. If m, n are different numbers, then one, and (by 83, 84) only one, of the chains m_0, n_0 is a proper part of the other; namely, either $n_0 \subset m'_0$ or $m_0 \subset n'_0$. **Proof.** If n is contained in m_0 , hence by 74 also $n_0 \subset m_0$, then m cannot be contained in the chain n_0 (for otherwise, by 74, also $m_0 \subset n_0$, hence $m_0 = n_0$, and thus by 84 also $m = n$), and from this it follows by 85 that $n_0 \subset m'_0$. In the contrary case, if n is not contained in the chain m_0 , then by 85 $m_0 \subset n'_0$, as was to be proved.

89. Definition. The number m is called smaller than the number n , and at the same time n is called greater than m , in signs

$$m < n \quad \text{and} \quad n > m,$$

if the condition

$$n_0 \subset m'_0$$

is fulfilled; by 74 this condition can also be expressed by

$$n \subset m'_0.$$

90. Proposition. If m, n are any numbers, then one and only one of the following cases λ, μ, ν always occurs:

$$\lambda. \quad m = n, \quad n = m, \quad \text{that is, } m_0 = n_0,$$

$$\mu. \quad m < n, \quad n > m, \quad \text{that is, } n_0 \subset m'_0,$$

$$\nu. \quad m > n, \quad n < m, \quad \text{that is, } m_0 \subset n'_0.$$

Proof. For if λ occurs (84), neither μ nor ν can occur, because by 83 one never has $n_0 \subset n'_0$. But if λ does not occur, then by 88 one and only one of the cases μ, ν occurs, as was to be proved.

91. Proposition. One has $n < n'$. **Proof.** For the condition for case ν in 90 is fulfilled by $m = n'$.

92. Definition. To express that m is either $= n$ or $< n$, and hence not $> n$ (90), one uses the notation

$$m \leq n \quad \text{or also} \quad n \geq m,$$

and says that m is at most equal to n , and n at least equal to m .

93. Proposition. Each of the conditions

$$m \leq n, \quad m < n', \quad n_0 \subset m_0$$

is equivalent to each of the others. **Proof.** For if $m \leq n$, then from λ, μ in 90 it always follows that $n_0 \subset m_0$, because (by 76) $m'_0 \subset m_0$. Conversely, if $n_0 \subset m_0$, hence by 74 also $n \subset m_0$, then from $m_0 = \mathfrak{M}(m, m'_0)$ it follows that either $n = m$ or $n \subset m'_0$, that is, $n > m$. Thus the condition $m \leq n$ is equivalent to $n_0 \subset m_0$. Furthermore, from 22, 27, and 75 it follows that this condition $n_0 \subset m_0$ is again equivalent to $n'_0 \subset m'_0$, that is (by μ in 90), to $m < n'$, as was to be proved.

94. Proposition. Each of the conditions

$$m' \leq n, \quad m' < n', \quad m < n$$

is equivalent to each of the others. The proof follows immediately from 93, replacing m there by m' , and from μ in 90.

95. Proposition. If $l < m$ and $m \leq n$, or if $l \leq m$ and $m < n$, then $l < n$. But if $l \leq m$ and $m \leq n$, then $l \leq n$. **Proof.** For from the corresponding conditions (by 89 and 93) $m_0 \subset l'_0$ and $n_0 \subset m_0$ follows (by 7) $n_0 \subset l'_0$, and the same follows from the conditions $m_0 \subset l_0$ and $n_0 \subset m'_0$, because by the former also $m'_0 \subset l'_0$. Finally, from $m_0 \subset l_0$ and $n_0 \subset m_0$ follows also $n_0 \subset l_0$, as was to be proved.

96. Proposition. In every part T of N there is one and only one smallest number k , that is, a number k that is smaller than every other number contained in T . If T consists of a single number, then that same number is also the smallest number in T . **Proof.** Since T_0 is a chain (44), by 87 there is a number k whose chain $k_0 = T_0$. From this (by 45 and 77) it follows that $T \subset \mathfrak{M}(k, k'_0)$; hence first k itself must be contained in T (for otherwise $T \subset k'_0$, and therefore by 47 also $T_0 \subset k'_0$, that is, $k_0 \subset k'_0$, which is impossible by 83). Moreover, every number of the system T different from k must be contained in k'_0 , that is, must be $> k$ (89). At the same time, it follows by 90 that there is only one smallest number in T , as was to be proved.

97. Proposition. The smallest number of the chain n_0 is n , and the base-number 1 is the smallest of all numbers. **Proof.** For by 74 and 93 the condition $m \subset n_0$ is equivalent to $m \geq n$. Or the proposition also follows immediately from the proof of the preceding proposition, because if there $T = n_0$ is assumed, evidently $k = n$ (51).

98. Definition. If n is any number, then by Z_n we shall denote the system of all numbers that are not greater than n , and hence are not contained in n'_0 . By 92 and 93 the condition

$$m \subset Z_n$$

is evidently equivalent to each of the following conditions:

$$m \leq n, \quad m < n', \quad n_0 \subset m_0.$$

99. Proposition. $1 \subset Z_n$ and $n \subset Z_n$. The proof follows from 98, or also from 71 and 82.

100. Proposition. Each of the conditions equivalent by 98,

$$m \subset Z_n, \quad m \leq n, \quad m < n', \quad n_0 \subset m_0,$$

is also equivalent to the condition

$$Z_m \subset Z_n.$$

Proof. For if $m \subset Z_n$, hence $m \leq n$, and if $l \subset Z_m$, hence $l \leq m$, then by 95 also $l \leq n$, that is, $l \subset Z_n$; hence if $m \subset Z_n$, every element l of the system Z_m is also an element of Z_n , that is, $Z_m \subset Z_n$. Conversely, if $Z_m \subset Z_n$, then by 7 also $m \subset Z_n$, because (by 99) $m \subset Z_m$, as was to be proved.

101. Proposition. The conditions for the cases λ, μ, ν in 90 can also be represented in the following way:

$$\begin{aligned}\lambda. \quad & m = n, \quad n = m, \quad Z_m = Z_n, \\ \mu. \quad & m < n, \quad n > m, \quad Z_m \text{ proper part of } Z_n, \\ \nu. \quad & m > n, \quad n < m, \quad Z_n \text{ proper part of } Z_m.\end{aligned}$$

The proof follows immediately from 90, if one considers that by 100 the conditions $n_0 \subset m_0$ and $Z_m \subset Z_n$ are equivalent.

102. Proposition. $Z_1 = 1$. **Proof.** For the base-number 1 is contained in Z_1 by 99, and every number different from 1 is in $1'_0$ by 78, hence not contained in Z_1 by 98, as was to be proved.

103. Proposition. By 98,

$$N = \mathfrak{M}(Z_n, n'_0).$$

104. Proposition. One has

$$n = \mathfrak{G}(Z_n, n_0),$$

that is, n is the only common element of the systems Z_n and n_0 . **Proof.** From 99 and 74 it follows that n is contained in Z_n and n_0 ; but every element of the chain n_0 different from n is by 77 in n'_0 , hence by 98 not contained in Z_n , as was to be proved.

105. Proposition. By 91 and 98, the number n' is not contained in Z_n .

106. Proposition. If $m < n$, then Z_m is a proper part of Z_n , and conversely. **Proof.** If $m < n$, then (by 100) $Z_m \subset Z_n$; and since the number n contained in Z_n by 99 cannot by 98 be contained in Z_m , because $n > m$, Z_m is a proper part of Z_n . Conversely, if Z_m is a proper part of Z_n , then (by 100) $m \leq n$; and since m cannot be $= n$, because otherwise also $Z_m = Z_n$, it must be that $m < n$, as was to be proved.

107. Proposition. Z_n is a proper part of $Z_{n'}$. The proof follows from 106, since (by 91) $n < n'$.

108. Proposition. $Z_{n'} = \mathfrak{M}(Z_n, n')$. **Proof.** For every number contained in $Z_{n'}$ is (by 98) $\leq n'$, hence is either $= n'$ or $< n'$, and in the latter case, by 98, an element of Z_n ; hence certainly $Z_{n'} \subset \mathfrak{M}(Z_n, n')$. Conversely, since (by 107) $Z_n \subset Z_{n'}$ and (by 99) $n' \in Z_{n'}$, by 10

$$\mathfrak{M}(Z_n, n') \subset Z_{n'}$$

follows; hence our proposition follows by 5.

109. Proposition. The image Z'_n of the system Z_n is a proper part of the system $Z_{n'}$. **Proof.** For every number contained in Z'_n is the image m' of a number m contained in Z_n , and since $m \leq n$, hence (by 94) $m' \leq n'$, it follows (by 98) that $Z'_n \subset Z_{n'}$. Since, moreover, the number 1 is contained in $Z_{n'}$ by 99, but cannot by 71 be contained in the image Z'_n , Z'_n is a proper part of $Z_{n'}$, as was to be proved.

110. Proposition. $Z_{n'} = \mathfrak{M}(1, Z'_n)$. **Proof.** Every number of the system $Z_{n'}$ different from 1 is by 78 the image m' of a number m , and this must be $\leq n$, hence contained in Z_n by 98 (for otherwise $m > n$, hence by 94 also $m' > n'$, and so by 98 m' would not be contained in $Z_{n'}$); from $m \in Z_n$ follows $m' \in Z'_n$, and consequently certainly

$$Z_{n'} \subset \mathfrak{M}(1, Z'_n).$$

Conversely, since (by 99) $1 \in Z_{n'}$ and (by 109) $Z'_n \subset Z_{n'}$, by 10 $\mathfrak{M}(1, Z'_n) \subset Z_{n'}$, and from this our proposition follows by 5.

111. Definition. If in a system E of numbers there is an element g that is greater than every other number contained in E , then g is called the greatest number of

the system E ; and evidently, by 90, there can be only one such greatest number in E . If a system consists of a single number, then that same number is the greatest number of the system.

112. Proposition. By 98, n is the greatest number of the system Z_n .

113. Proposition. If there is in E a greatest number g , then $E \subset Z_g$. **Proof.** For every number contained in E is $\leq g$, hence by 98 contained in Z_g , as was to be proved.

114. Proposition. If E is part of a system Z_n , or, what says the same thing, if there is a number n such that all numbers contained in E are $\leq n$, then E has a greatest number g . **Proof.** The system of all numbers p satisfying the condition $E \subset Z_p$ — and by our assumption such numbers exist — is a chain (37), because by 107 and 7 also $E \subset Z_{p'}$ follows, and therefore it is (by 87) $= g_0$, where g denotes the smallest of these numbers (96, 97). Thus also $E \subset Z_g$; consequently (98) every number contained in E is $\leq g$, and we still have only to show that the number g itself is contained in E . This is immediately evident if $g = 1$, because then (by 102) Z_g , and consequently also E , consists of the single number 1. But if g is different from 1 and therefore by 78 is the image f' of a number f , then (by 108) $E \subset \mathfrak{M}(Z_f, g)$; if now g were not contained in E , then it would have to be that $E \subset Z_f$, and hence among the numbers p there would be a number f which (by 91) is $< g$, contrary to what was established above. Thus g is contained in E , as was to be proved.

115. Definition. If $l < m$ and $m < n$, then we say that the number m lies between l and n (also between n and l).

116. Proposition. There is no number that lies between n and n' . **Proof.** For as soon as $m < n'$, and hence (by 93) $m \leq n$, by 90 it cannot be that $n < m$, as was to be proved.

117. Proposition. If t is a number in T , but not the smallest (96), then in T there is one and only one next smaller number s , that is, a number s such that $s < t$ and such that in T there is no number lying between s and t . Likewise, if t is not perhaps the greatest number in T (111), then in T there is always one and only one next greater number u , that is, a number u such that $t < u$ and such that in T there is no number lying between t and u . At the same time, t is in T next greater than s and next smaller than u .

Proof. If t is not the smallest number in T , let E be the system of all those numbers of T that are $< t$; then (by 98) $E \subset Z_t$, and consequently (114) there is in E a greatest number s , which evidently has the properties stated in the proposition and is also the only such number. Furthermore, if t is not the greatest number in T , then by 96, among all the numbers of T that are $> t$, there is certainly a smallest number u , which, and which alone, has the properties stated in the proposition. The correctness of the final remark of the proposition is likewise evident.

118. Proposition. In N , the number n' is next greater than n , and n next smaller than n' .

The proof follows from 116 and 117.

§ 8. Finite and infinite parts of the sequence of numbers

119. Proposition. Every system Z_n in 98 is finite.

Proof by complete induction (80). For

$\rho.$ the proposition is true for $n = 1$ by 65 and 102.

σ . If Z_n is finite, then by 108 and 70 $Z_{n'}$ is also finite, as was to be proved.

120. Proposition. If m, n are different numbers, then Z_m, Z_n are non-similar systems.

Proof. For symmetry's sake we may by 90 assume that $m < n$; then by 106 Z_m is a proper part of Z_n , and since by 119 Z_n is finite, by 64 Z_m and Z_n cannot be similar, as was to be proved.

121. Proposition. Every part E of the sequence of numbers N that has a greatest number (111) is finite.

The proof follows from 113, 119, and 68.

122. Proposition. Every part U of the sequence of numbers N that has no greatest number is simply infinite (71).

Proof. If u is any number in U , then by 117 there is in U one and only one next greater number than u , which we shall denote by $\psi(u)$ and regard as the image of u . The mapping ψ of the system U that is completely determined in this way evidently has the property

$$\alpha. \quad \psi(U) \subset U,$$

that is, U is mapped by ψ into itself. If, further, u, v are different numbers in U , then for symmetry's sake we may by 90 assume that $u < v$; then from the definition of ψ and by 117 it follows that $\psi(u) \leq v$ and $v < \psi(v)$, hence by 95 $\psi(u) < \psi(v)$. Therefore by 90 the images $\psi(u), \psi(v)$ are different, that is,

δ . the mapping ψ is similar.

Furthermore, let u_1 denote the smallest number (96) of the system U . Then every number u contained in U is $\geq u_1$, and since generally $u < \psi(u)$, by 95 $u_1 < \psi(u)$; hence by 90 u_1 is different from $\psi(u)$, that is,

γ . the element u_1 of U is not contained in $\psi(U)$.

Thus $\psi(U)$ is a proper part of U , and consequently U is an infinite system by 64. Now, in agreement with 44, if V is any part of U , let $\psi_0(V)$ denote the chain of V corresponding to the mapping ψ ; we must finally still show that

$$\beta. \quad U = \psi_0(u_1)$$

holds. Indeed, since every such chain $\psi_0(V)$ is, by its definition (44), part of the system U mapped by ψ into itself, it is self-evident that $\psi_0(u_1) \subset U$. Conversely, it follows at once from 45 that the element u_1 contained in U is certainly contained in $\psi_0(u_1)$. But suppose that there were elements of U not contained in $\psi_0(u_1)$. Then by 96 there would have to be among them a smallest number w ; and since, by what has just been said, this is different from the smallest number u_1 of the system U , by 117 there must also be in U a number v that is next smaller than w , from which it follows at the same time that $w = \psi(v)$. Since now $v < w$, by the definition of w the number v must certainly be contained in $\psi_0(u_1)$; from this, however, by 55 it follows that also $\psi(v)$, hence w , must be contained in $\psi_0(u_1)$, and since this contradicts the definition of w , our preceding supposition is inadmissible. Hence $U \subset \psi_0(u_1)$ and consequently also $U = \psi_0(u_1)$, as asserted. From $\alpha, \beta, \gamma, \delta$ it follows now by 71 that U is a simply infinite system ordered by ψ , as was to be proved.

123. Proposition. In consequence of 121 and 122, any part T of the sequence of numbers N is finite or simply infinite according as there is, or is not, a greatest number in T .

§ 9. Definition of a mapping of the sequence of numbers by induction

124. In what follows we shall also denote numbers by small Latin letters and shall keep, in general, all notation from the preceding §§ 6 to 8, while Ω denotes an arbitrary system whose elements need not be contained in N .

125. Proposition. If an arbitrary (similar or non-similar) mapping θ of a system Ω into itself is given, and in addition a definite element ω in Ω is given, then to every number n there corresponds one and only one mapping ψ_n of the corresponding number-system Z_n defined in 98, satisfying the conditions¹⁴

- I. $\psi_n(Z_n) \subset \Omega$,
- II. $\psi_n(1) = \omega$,
- III. $\psi_n(t') = \theta\psi_n(t)$, if $t < n$,

where the sign $\theta\psi_n$ has the meaning given in 25.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$. In this case, by 102, the system Z_n consists of the single number 1, and therefore the mapping ψ_1 is already completely, and in such a way, defined by II that I is fulfilled, while III is wholly absent.

σ . If the proposition is true for a number n , we show that it is also true for the following number $p = n'$, and we begin by proving that there can be only one corresponding mapping ψ_p of the system Z_p . Indeed, suppose that a mapping ψ_p satisfies the conditions

- I'. $\psi_p(Z_p) \subset \Omega$,
- II'. $\psi_p(1) = \omega$,
- III'. $\psi_p(m') = \theta\psi_p(m)$, if $m < p$.

Then, since $Z_n \subset Z_p$ by 107, this mapping contains, by 21, also a mapping of Z_n , which evidently satisfies the same conditions I, II, III as ψ_n and consequently agrees entirely with ψ_n ; hence for all numbers m contained in Z_n , that is, by 98, for all numbers m that are $< p$, i.e. $\leq n$, one must have

$$\psi_p(m) = \psi_n(m) \quad (m)$$

and as a special case also

$$\psi_p(n) = \psi_n(n). \quad (n)$$

Since, furthermore, by 105 and 108, p is the only number of the system Z_p not contained in Z_n , and since by III' and (n) also

$$\psi_p(p) = \theta\psi_n(n) \quad (p)$$

must hold, the correctness of our assertion follows: there can be only one mapping ψ_p of the system Z_p satisfying I', II', III', because ψ_p has been completely reduced to ψ_n by the just derived conditions (m) and (p). We have now to show conversely that this mapping ψ_p of the system Z_p , completely determined by (m) and (p), really satisfies I', II', III'. Evidently I' follows from (m) and (p), with regard to I and to the

¹⁴For clarity I have here and in the following proposition 126 stated condition I separately, although it is really already a consequence of II and III.

fact that $\theta(\Omega) \subset \Omega$. Likewise II' follows from (m) and II, since by 99 the number 1 is contained in Z_n . The correctness of III' follows first for those numbers m that are $< n$ from (m) and III, and for the only remaining number $m = n$ it follows from (p) and (n). This proves completely that from the validity of our proposition for the number n there always follows its validity for the following number p , as was to be proved.

126. Proposition of definition by induction. If an arbitrary (similar or non-similar) mapping θ of a system Ω into itself is given, and in addition a definite element ω in Ω is given, then there is one and only one mapping ψ of the sequence of numbers N satisfying the conditions

- I. $\psi(N) \subset \Omega$,
- II. $\psi(1) = \omega$,
- III. $\psi(n') = \theta\psi(n)$,

where n denotes any number.

Proof. If such a mapping ψ really exists, then by 21 it contains a mapping ψ_n of the system Z_n satisfying the conditions I, II, III stated in 125. Since there is always one and only one such mapping ψ_n , necessarily

$$\psi(n) = \psi_n(n) \tag{n}$$

must hold. Since ψ is thereby completely determined, it follows that there can be only one such mapping ψ (cf. the end of 130). Conversely, that the mapping ψ determined by (n) also satisfies our conditions I, II, III follows easily from (n), taking account of the properties I, II, and (p) proved in 125, as was to be proved.

127. Proposition. Under the hypotheses made in the preceding proposition,

$$\psi(T') = \theta\psi(T),$$

where T denotes any part of the sequence of numbers N .

Proof. For if t denotes every number of the system T , then $\psi(T')$ consists of all elements $\psi(t')$, and $\theta\psi(T)$ consists of all elements $\theta\psi(t)$; our proposition follows from this, since by III in 126, $\psi(t') = \theta\psi(t)$.

128. Proposition. Keeping the same hypotheses, and denoting by θ_0 the chains (44) that correspond to the mapping θ of the system Ω into itself, one has

$$\psi(N) = \theta_0(\omega).$$

Proof. First we show by complete induction (80) that

$$\psi(N) \subset \theta_0(\omega),$$

that is, that every image $\psi(n)$ is also an element of $\theta_0(\omega)$. Indeed,

ρ . this proposition is true for $n = 1$, because by 126. II $\psi(1) = \omega$, and by 45 $\omega \subset \theta_0(\omega)$.
 σ . If the proposition is true for a number n , that is, if $\psi(n) \subset \theta_0(\omega)$, then by 55 also $\theta(\psi(n)) \subset \theta_0(\omega)$, that is, by 126. III, $\psi(n') \subset \theta_0(\omega)$; hence the proposition is also true for the following number n' , as was to be proved.

In order next to prove that every element v of the chain $\theta_0(\omega)$ is contained in $\psi(N)$, that is, that

$$\theta_0(\omega) \subset \psi(N),$$

we again apply complete induction, namely proposition 59 transferred to Ω and the mapping θ . Indeed,

ρ . the element ω is $= \psi(1)$, and hence is contained in $\psi(N)$.

σ . If v is a common element of the chain $\theta_0(\omega)$ and of the system $\psi(N)$, then $v = \psi(n)$, where n denotes a number; from this it follows by 126. III that $\theta(v) = \theta\psi(n) = \psi(n')$, so that $\theta(v)$ is also contained in $\psi(N)$, as was to be proved.

From the proved propositions $\psi(N) \subset \theta_0(\omega)$ and $\theta_0(\omega) \subset \psi(N)$ it follows by 5 that $\psi(N) = \theta_0(\omega)$, as was to be proved.

129. Proposition. Under the same hypotheses, generally

$$\psi(n_0) = \theta_0(\psi(n)).$$

Proof by complete induction 80. For

ρ . the proposition holds by 128 for $n = 1$, since $1_0 = N$ and $\psi(1) = \omega$.

σ . If the proposition is true for a number n , then

$$\theta(\psi(n_0)) = \theta(\theta_0(\psi(n)));$$

now by 127 and 75

$$\theta(\psi(n_0)) = \psi(n'_0),$$

and by 57 and 126. III

$$\theta(\theta_0(\psi(n))) = \theta_0(\theta(\psi(n))) = \theta_0(\psi(n'));$$

hence

$$\psi(n'_0) = \theta_0(\psi(n')),$$

that is, the proposition is also true for the number n' following n , as was to be proved.

130. Remark. Before passing to the most important applications of proposition 126, proved above, on definition by induction (§§ 10 to 14), it is worth the trouble to call attention to a circumstance by which it differs essentially from the proposition of demonstration by induction proved in 80, or rather already in 59, 60, however close the kinship between the former and the latter may seem. Namely, while proposition 59 holds quite generally for every chain A_0 , where A is any part of a system S mapped into itself by an arbitrary mapping φ (§ 4), the matter is quite different with proposition 126, which asserts only the existence of a contradiction-free, or single-valued, mapping ψ of the simply infinite system 1_0 . If in the latter proposition, while retaining the hypotheses on Ω and θ , one wished to put in place of the sequence of numbers 1_0 an arbitrary chain A_0 from such a system S , and to define a mapping ψ of A_0 into Ω in a manner similar to 126. II, III, by requiring that

ρ . to every element a of A there correspond a definite element $\psi(a)$ chosen from Ω , and

σ . for every element n contained in A_0 and its image $n' = \varphi(n)$, the condition $\psi(n') = \theta\psi(n)$ should hold,

then very often the case would occur that no such mapping ψ exists at all, because these conditions ρ, σ can come into contradiction with one another even if the freedom of choice contained in ρ is restricted in advance in accordance with condition σ . An example will suffice to become convinced of this. Let the system S consisting of the two different elements a and b be mapped by φ into itself in such a way that

$a' = b, b' = a$; then evidently $a_0 = b_0 = S$. Further, let the system Ω consisting of the three different elements α, β, γ be mapped by θ into itself so that $\theta(\alpha) = \beta, \theta(\beta) = \gamma, \theta(\gamma) = \alpha$. If one now requires a mapping ψ of a_0 into Ω such that $\psi(a) = \alpha$ and in addition, for every element n contained in a_0 , always $\psi(n') = \theta\psi(n)$, then one meets a contradiction; for with $n = a$ one obtains $\psi(b) = \theta(\alpha) = \beta$, and from this, with $n = b$, it would follow that $\psi(a) = \theta(\beta) = \gamma$, whereas $\psi(a) = \alpha$ had been required.

But if there is a mapping ψ of A_0 into Ω satisfying the above conditions ρ, σ without contradiction, then it follows easily from 60 that it is completely determined. For if the mapping χ satisfies the same conditions, then generally $\chi(n) = \psi(n)$, because by ρ this proposition holds for all elements $n = a$ contained in A , and because, if it holds for an element n of A_0 , then by σ it must also hold for its image n' .

131. In order to bring to light the scope of our proposition 126, we insert here a consideration that is also useful for other investigations, for example for the so-called theory of groups.

We consider a system Ω whose elements admit a certain composition in such a way that from an element ν , by the action of an element ω , there always arises again a definite element of the same system Ω , which may be denoted by $\omega \cdot \nu$ or $\omega\nu$, and is in general to be distinguished from $\nu\omega$. One may also understand this by saying that to every definite element ω there corresponds a definite mapping of the system Ω into itself, perhaps denoted by ω , insofar as every element ν yields the definite image $\omega(\nu) = \omega\nu$. Applying proposition 126 to this system Ω and its element ω , while at the same time replacing the mapping there denoted by θ with ω , to every number n there corresponds a definite element $\psi(n)$ contained in Ω , which may now be denoted by the symbol ω^n and is sometimes called the n -th power of ω . This concept is completely explained by the conditions imposed on it,

$$\text{II. } \omega^1 = \omega, \quad \text{III. } \omega^{n'} = \omega\omega^n,$$

and its existence is secured by the proof of proposition 126.

If the preceding composition of elements is, in addition, such that for arbitrary elements μ, ν, ω one always has $\omega(\nu\mu) = (\omega\nu)\mu$, then also the propositions

$$\omega^{n'} = \omega^n\omega, \quad \omega^m\omega^n = \omega^n\omega^m$$

hold; their proofs may easily be carried out by complete induction (80), and may be left to the reader.

The preceding general consideration can be applied immediately to the following example. If S is a system of arbitrary elements, and Ω the corresponding system whose elements are all mappings ν of S into itself (36), then these elements can always be composed by 25, because $\nu(S) \subset S$, and the mapping $\omega\nu$ composed from such mappings ν and ω is itself again an element of Ω . Then all elements ω^n are also mappings of S into itself, and one says that they arise by repetition of the mapping ω . We shall now point out a simple connection between this concept and the concept of the chain $\omega_0(A)$ explained in 44, where A again denotes any part of S . For brevity, let the image $\omega^n(A)$ produced by the mapping ω^n be denoted by A_n ; then it follows from III and 25 that $\omega(A_n) = A_{n'}$. From this it follows easily by complete induction (80) that all these systems A_n are parts of the chain $\omega_0(A)$; for ρ . this assertion holds by 50 for $n = 1$, and

σ . if it holds for a number n , then from 55 and from $A_{n'} = \omega(A_n)$ it follows that it also holds for the following n' , as was to be proved. Since further, by 45, $A \subset \omega_0(A)$, it follows from 10 that also the system K composed of A and all the images A_n

is part of $\omega_0(A)$. Conversely, since by 23 $\omega(K)$ is composed of $\omega(A) = A_1$ and all systems $\omega(A_n) = A_{n'}$, hence by 78 of all systems A_n , which by 9 are parts of K , it follows by 10 that $\omega(K) \subset K$, that is, K is a chain (37); and since by 9 $A \subset K$, by 47 also $\omega_0(A) \subset K$. Hence $\omega_0(A) = K$, that is, the following proposition holds: if ω is a mapping of a system S into itself and A is any part of S , then the chain of A corresponding to the mapping ω is composed of A and all images $\omega^n(A)$ arising by repetition of ω . We recommend that the reader return to the earlier propositions 57, 58 with this conception of a chain.

§ 10. The class of simply infinite systems

132. Proposition. All simply infinite systems are similar to the sequence of numbers N , and consequently by 33 also to one another.

Proof. Let the simply infinite system Ω be ordered by the mapping θ (71), and let ω be the fundamental element of Ω occurring here. If, again, we denote by θ_0 the chains corresponding to the mapping θ (44), then by 71 the following hold:

- $\alpha.$ $\theta(\Omega) \subset \Omega$,
- $\beta.$ $\Omega = \theta_0(\omega)$,
- $\gamma.$ ω is not contained in $\theta(\Omega)$,
- $\delta.$ the mapping θ is similar.

Let ψ denote the mapping of the sequence of numbers N defined in 126. Then from β and 128 first follows

$$\psi(N) = \Omega,$$

and therefore by 32 we have only to show that ψ is a similar mapping, that is, by 26, that to different numbers m, n there correspond different images $\psi(m), \psi(n)$. For symmetry's sake we may by 90 assume that $m > n$, hence $m \subset n'_0$, and the proposition to be proved amounts to saying that $\psi(n)$ is not contained in $\psi(n'_0)$, and therefore by 127 not contained in $\theta\psi(n_0)$. We prove this for every number n by complete induction (80). Indeed,

- $\rho.$ this proposition holds by γ for $n = 1$, because $\psi(1) = \omega$ and $\psi(1_0) = \psi(N) = \Omega$.
- $\sigma.$ If the proposition is true for a number n , then it is also true for the following number n' ; for if $\psi(n')$, that is, $\theta\psi(n)$, were contained in $\theta\psi(n'_0)$, then by δ and 27, $\psi(n)$ would also have to be contained in $\psi(n'_0)$, whereas our assumption says exactly the contrary, as was to be proved.

133. Proposition. Every system that is similar to a simply infinite system, and consequently by 132 and 33 also to the sequence of numbers N , is simply infinite.

Proof. If Ω is a system similar to the sequence of numbers N , then by 32 there is a similar mapping ψ of N such that

$$\text{I. } \psi(N) = \Omega$$

and we set

$$\text{II. } \psi(1) = \omega.$$

If by 26 $\bar{\psi}$ denotes the inverse, likewise similar, mapping of Ω , then to every element ν of Ω there corresponds a definite number $\bar{\psi}(\nu) = n$, namely the one whose image is $\psi(n) = \nu$. Since to this number n there corresponds a definite following number $\varphi(n) = n'$, and to this again a definite element $\psi(n')$ in Ω , to every element ν of the

system Ω there belongs a definite element $\psi(n')$ of the same system, which we shall denote as the image of ν by $\theta(\nu)$. This completely determines a mapping θ of Ω into itself¹⁵; and to prove our proposition we shall show that Ω is ordered by θ as a simply infinite system (71), that is, that all the conditions $\alpha, \beta, \gamma, \delta$ stated in the proof of 132 are fulfilled. First α is immediately clear from the definition of θ . Since further to every number n there corresponds an element $\nu = \psi(n)$ for which $\theta(\nu) = \psi(n')$, one has generally

$$\text{III. } \psi(n') = \theta\psi(n),$$

and from this, in connection with I, II, α , it follows that the mappings θ, ψ satisfy all conditions of proposition 126; hence β follows from 128 and I. By 127 and I, furthermore,

$$\psi(N') = \theta\psi(N) = \theta(\Omega),$$

and from this, in connection with II and the similarity of ψ, γ follows, because otherwise $\psi(1)$ would be contained in $\psi(N')$, and hence by 27 the number 1 would have to be contained in N' , which by 71. γ is not the case. Finally, if μ, ν are elements of Ω , and m, n denote the corresponding numbers whose images are $\psi(m) = \mu, \psi(n) = \nu$, then from the assumption $\theta(\mu) = \theta(\nu)$ it follows, by what has just been said, that $\psi(m') = \psi(n')$; from this, because ψ, φ are similar, it follows that $m' = n', m = n$, and therefore also $\mu = \nu$. Hence δ also holds, as was to be proved.

134. Remark. In consequence of the two preceding propositions 132, 133, all simply infinite systems form a class in the sense of 34. At the same time, with regard to 71 and 73, it is clear that every proposition about the numbers, that is, about the elements n of the simply infinite system N ordered by the mapping φ , and indeed every such proposition in which the special nature of the elements n is wholly disregarded and only concepts arising from the ordering φ are involved, possesses completely general validity also for every other simply infinite system Ω ordered by a mapping θ and for its elements ν . The transfer from N to Ω (for example, also the translation of an arithmetical proposition from one language into another) is effected by the mapping ψ considered in 132, 133, which transforms every element n of N into an element ν of Ω , namely into $\psi(n)$. This element ν may be called the n -th element of Ω , and accordingly the number n itself is the n -th number of the sequence of numbers N . The same significance that the mapping φ has for the laws in the domain N , insofar as to every element n there follows a definite element $\varphi(n) = n'$, belongs, after the transformation effected by ψ , to the mapping θ for the same laws in the domain Ω , insofar as to the element $\nu = \psi(n)$ produced by transformation of n there follows the element $\theta(\nu) = \psi(n')$ produced by transformation of n' . Thus one may rightly say that φ is transformed by ψ into θ , which is expressed symbolically by $\theta = \psi\varphi\bar{\psi}, \varphi = \bar{\psi}\theta\psi$. By these remarks, I believe, the explanation of the concept of numbers given in 73 is completely justified. We now pass to further applications of proposition 126.

§ 11. Addition of numbers

135. Definition. It is natural to apply the definition, presented in proposition 126, of a mapping ψ of the sequence of numbers N , or of the function $\psi(n)$ thereby determined, to the case in which the system there denoted by Ω , in which the image $\psi(N)$ is to be contained, is the sequence of numbers N itself; for this system

¹⁵Evidently θ is the mapping $\psi\varphi\bar{\psi}$ composed from $\bar{\psi}, \varphi, \psi$ according to 25.

Ω already has a mapping θ of Ω into itself, namely the mapping φ by which N is ordered as a simply infinite system (71, 73). Then $\Omega = N$, $\theta(n) = \varphi(n) = n'$, and hence

$$\text{I. } \psi(N) \subset N.$$

It remains, in order to determine ψ completely, only to choose the element ω from Ω , that is, from N , at will. If we take $\omega = 1$, then ψ is evidently the identical mapping (21) of N , because the conditions

$$\psi(1) = 1, \quad \psi(n') = (\psi(n))'$$

are generally satisfied by $\psi(n) = n$. If, therefore, another mapping ψ of N is to be generated, then for ω one must choose a number m' different from 1 and contained by 78 in N' , where m itself denotes an arbitrary number. Since the mapping ψ evidently depends on the choice of this number m , we denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol $m + n$ and call this number the sum that arises from the number m by addition of the number n , or briefly the sum of the numbers m, n . It is therefore completely determined by 126 by the conditions¹⁶

$$\text{II. } m + 1 = m', \quad \text{III. } m + n' = (m + n)'.$$

136. Proposition. One has $m' + n = m + n'$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$, because by 135. II

$$m' + 1 = (m')' = (m + 1)'$$

and by 135. III, $(m + 1)' = m + 1'$.

σ . If the proposition is true for a number n , and if one sets the following number $n' = p$, then $m' + n = m + p$, hence also $(m' + n)' = (m + p)'$; from this, by 135. III, $m' + p = m + p'$ follows. Thus the proposition holds also for the following number p , as was to be proved.

137. Proposition. One has $m' + n = (m + n)'$.

The proof follows from 136 and 135. III.

138. Proposition. One has $1 + n = n'$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II for $n = 1$.

σ . If the proposition holds for a number n , and if $n' = p$, then $1 + n = p$, hence also $(1 + n)' = p'$; therefore by 135. III, $1 + p = p'$, that is, the proposition holds also for the following number p , as was to be proved.

139. Proposition. One has $1 + n = n + 1$.

The proof follows from 138 and 135. II.

140. Proposition. One has $m + n = n + m$.

Proof by complete induction (80). For

ρ . the proposition is true by 139 for $n = 1$.

¹⁶The preceding definition of addition, founded directly on proposition 126, seems to me the simplest. With the aid of the concept developed in 131, however, one can also define the sum $m + n$ by $\varphi^n(m)$ or also by $\varphi^m(n)$, where φ again has the above meaning. To prove the complete agreement of these definitions with the preceding one, it is only necessary, by 126, to show that, if $\varphi^n(m)$ or $\varphi^m(n)$ is denoted by $\psi(n)$, the conditions $\psi(1) = m'$, $\psi(n') = \varphi\psi(n)$ are fulfilled; this is easily done with the aid of complete induction (80), using 131.

σ . If the proposition holds for a number n , then from it also follows $(m+n)' = (n+m)'$, that is, by 135. III, $m+n' = n+m'$; hence by 136, $m+n' = n'+m$. Thus the proposition holds also for the following number n' , as was to be proved.

141. Proposition. One has $(l+m) + n = l + (m+n)$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$, because by 135. II, III, II, $(l+m) + 1 = (l+m)' = l+m' = l + (m+1)$.

σ . If the proposition holds for a number n , then from it also follows $((l+m) + n)' = (l + (m+n))'$, that is, by 135. III,

$$(l+m) + n' = l + (m+n)' = l + (m+n'),$$

so the proposition holds also for the following number n' , as was to be proved.

142. Proposition. One has $m+n > m$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II and 91 for $n = 1$.

σ . If the proposition holds for a number n , then by 95 it also holds for the following number n' , because by 135. III and 91

$$m+n' = (m+n)' > m+n,$$

as was to be proved.

143. Proposition. The conditions $m > a$ and $m+n > a+n$ are equivalent.

Proof by complete induction (80). For

ρ . the proposition holds by 135. II and 94 for $n = 1$.

σ . If the proposition holds for a number n , then it also holds for the following number n' , because by 94 the condition $m+n > a+n$ is equivalent to $(m+n)' > (a+n)'$, and therefore by 135. III also to

$$m+n' > a+n',$$

as was to be proved.

144. Proposition. If $m > a$ and $n > b$, then also

$$m+n > a+b.$$

Proof. From our hypotheses it follows by 143 that $m+n > a+n$ and $n+a > b+a$, or, what is the same by 140, $a+n > a+b$; from this the proposition follows by 95.

145. Proposition. If $m+n = a+n$, then $m = a$.

Proof. For if m is not $= a$, then by 90 either $m > a$ or $m < a$; correspondingly, by 143, $m+n > a+n$ or $m+n < a+n$. Hence by 90, $m+n$ certainly cannot be $= a+n$, as was to be proved.

146. Proposition. If $l > n$, then there is one and, by 145, only one number m satisfying the condition $m+n = l$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$. Indeed, if $l > 1$, that is, by 89, if l is contained in N' and hence is the image m' of a number m , then by 135. II it follows that $l = m+1$, as was to be proved.

σ . If the proposition holds for a number n , we show that it also holds for the following number n' . Indeed, if $l > n'$, then by 91 and 95 also $l > n$, and consequently there is a number k satisfying $l = k+n$. Since by 138 this number is different from 1 (for otherwise $l = n'$), it is by 78 the image m' of a number m , and consequently $l = m'+n$, hence by 136 also $l = m+n'$, as was to be proved.

§ 12. Multiplication of numbers

147. Definition. After in the preceding § 11 an infinite system of new mappings of the sequence of numbers N into itself has been found, each of these may again be used, by 126, to generate further new mappings ψ of N . If there we put $\Omega = N$ and $\theta(n) = m + n = n + m$, where m is a definite number, then again certainly

$$\text{I. } \psi(N) \subset N,$$

and in order to determine ψ completely it remains only to choose the element ω from N at will. The simplest case occurs when this choice is brought into a certain agreement with the choice of θ , by putting $\omega = m$. Since the mapping ψ thereby completely determined depends on this number m , we denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol $m \times n$, or $m \cdot n$, or mn , and call this number the product that arises from the number m by multiplication by the number n , or briefly the product of the numbers m, n . It is therefore completely determined by 126 by the conditions

$$\text{II. } m \cdot 1 = m, \quad \text{III. } mn' = mn + m.$$

148. Proposition. One has $m'n = mn + n$.

Proof by complete induction (80). For

ρ . the proposition is true by 147. II and 135. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$m'n + m' = (mn + n) + m',$$

and from this, by 147. III, 141, 140, 136, 141, 147. III,

$$\begin{aligned} m'n' &= mn + (n + m') = mn + (m' + n) \\ &= mn + (m + n') = (mn + m) + n' = mn' + n'; \end{aligned}$$

hence the proposition holds also for the following number n' , as was to be proved.

149. Proposition. One has $1 \cdot n = n$.

Proof by complete induction (80). For

ρ . the proposition is true by 147. II for $n = 1$.

σ . If the proposition holds for a number n , then $1 \cdot n + 1 = n + 1$, that is, by 147. III and 135. II, $1 \cdot n' = n'$; hence the proposition also holds for the following number n' , as was to be proved.

150. Proposition. One has $mn = nm$.

Proof by complete induction (80). For

ρ . the proposition holds by 147. II and 149 for $n = 1$.

σ . If the proposition holds for a number n , then

$$mn + m = nm + m,$$

that is, by 147. III and 148, $mn' = n'm$; hence the proposition holds also for the following number n' , as was to be proved.

151. Proposition. One has $l(m + n) = lm + ln$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II, 147. III, 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$l(m+n) + l = (lm + ln) + l;$$

by 147. III and 135. III, however,

$$l(m+n) + l = l(m+n)' = l(m+n'),$$

and by 141 and 147. III,

$$(lm + ln) + l = lm + (ln + l) = lm + ln'.$$

Hence $l(m+n') = lm + ln'$, that is, the proposition holds also for the following number n' , as was to be proved.

152. Proposition. One has $(m+n)l = ml + nl$.

The proof follows from 151 and 150.

153. Proposition. One has $(lm)n = l(mn)$.

Proof by complete induction (80). For

ρ . the proposition holds by 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$(lm)n + lm = l(mn) + lm,$$

that is, by 147. III, 151, 147. III,

$$(lm)n' = l(mn + m) = l(mn'),$$

so the proposition also holds for the following number n' , as was to be proved.

154. Remark. If in 147 no relation had been assumed between ω and θ , but instead $\omega = k$, $\theta(n) = m + n$ had been set, then by 126 a less simple mapping ψ of the sequence of numbers N would have arisen; for the number 1, $\psi(1) = k$, and for every other number, hence every number contained in the form n' , one would have $\psi(n') = mn + k$. For, as one easily convinces oneself with the aid of the preceding propositions, this fulfills the condition $\psi(n') = \theta\psi(n)$, that is, $\psi(n') = m + \psi(n)$, for all numbers n .

§ 13. Exponentiation of numbers

155. Definition. If in proposition 126 we again put $\Omega = N$, and further put $\omega = a$, $\theta(n) = an = na$, then there arises a mapping ψ of N which again satisfies the condition

$$\text{I. } \psi(N) \subset N.$$

We denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol a^n , and call this number a power of the base a , while n is called the exponent of this power of a . This concept is therefore completely determined by the conditions

$$\text{II. } a^1 = a, \quad \text{III. } a^{n'} = a \cdot a^n = a^n \cdot a.$$

156. Proposition. One has $a^{m+n} = a^m \cdot a^n$.

Proof by complete induction (80). For

ρ . the proposition holds by 135. II, 155. III, and 155. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$a^{m+n} \cdot a = (a^m \cdot a^n)a;$$

but by 155. III and 135. III,

$$a^{m+n} \cdot a = a^{(m+n)'} = a^{m+n'},$$

and by 153 and 155. III,

$$(a^m \cdot a^n)a = a^m(a^n \cdot a) = a^m \cdot a^{n'}.$$

Thus $a^{m+n'} = a^m \cdot a^{n'}$, that is, the proposition also holds for the following number n' , as was to be proved.

157. Proposition. One has $(a^m)^n = a^{mn}$.

Proof by complete induction (80). For

ρ . the proposition holds by 155. II and 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$(a^m)^n \cdot a^m = a^{mn} \cdot a^m;$$

but by 155. III,

$$(a^m)^n \cdot a^m = (a^m)^{n'},$$

and by 156 and 147. III,

$$a^{mn} \cdot a^m = a^{mn+m} = a^{mn'}.$$

Thus $(a^m)^{n'} = a^{mn'}$, that is, the proposition also holds for the following number n' , as was to be proved.

158. Proposition. One has $(ab)^n = a^n \cdot b^n$.

Proof by complete induction (80). For

ρ . the proposition holds by 155. II for $n = 1$.

σ . If the proposition holds for a number n , then by 150, 153, and 155. III also

$$(ab)^n \cdot a = a(a^n \cdot b^n) = (a \cdot a^n)b^n = a^{n'} \cdot b^n,$$

and from this

$$((ab)^n \cdot a)b = (a^{n'} \cdot b^n)b;$$

but by 153 and 155. III,

$$((ab)^n \cdot a)b = (ab)^n \cdot (ab) = (ab)^{n'},$$

and likewise

$$(a^{n'} \cdot b^n)b = a^{n'} \cdot (b^n \cdot b) = a^{n'} \cdot b^{n'}.$$

Thus $(ab)^{n'} = a^{n'} \cdot b^{n'}$, that is, the proposition also holds for the following number n' , as was to be proved.

§ 14. Number of elements of a finite system

159. Proposition. If Σ is an infinite system, then every one of the number systems Z_n explained in 98 can be similarly mapped into Σ (that is, is similar to a part of Σ), and conversely.

Proof. If Σ is infinite, then by 72 there is certainly a part T of Σ which is simply infinite, and hence by 132 is similar to the sequence of numbers N ; consequently, by 35, every system Z_n , as a part of N , is similar also to a part of T , and hence also to a part of Σ , as was to be proved.

The proof of the converse—however evident it may seem—is more complicated. If every system Z_n can be similarly mapped into Σ , then to every number n there corresponds such a similar mapping α_n of Z_n that $\alpha_n(Z_n) \subset \Sigma$. From the existence of such a series of mappings α_n , to be regarded as given but about which nothing further is assumed, we shall first derive, with the aid of proposition 126, the existence of a new series of such mappings ψ_n which has the special property that whenever $m \leq n$, and hence by 100 $Z_m \subset Z_n$, the mapping ψ_m of the part Z_m is contained in the mapping ψ_n of Z_n (21); that is, the mappings ψ_m and ψ_n agree completely with each other for all numbers contained in Z_m , and hence always also

$$\psi_m(m) = \psi_n(m).$$

In order to apply the cited proposition in accordance with this aim, let Ω mean the system whose elements are all possible similar mappings of all systems Z_n into Σ , and, with the aid of the given elements α_n , which are likewise contained in Ω , define in the following way a mapping θ of Ω into itself. If β is any element of Ω , hence, for example, a similar mapping of the definite system Z_n into Σ , then the system $\alpha_{n'}(Z_{n'})$ cannot be part of $\beta(Z_n)$; otherwise $Z_{n'}$ would, by 35, be similar to a part of Z_n , and hence by 107 to a proper part of itself, and consequently would be infinite, contrary to proposition 119. There are therefore certainly in $Z_{n'}$ one or several numbers p such that $\alpha_{n'}(p)$ is not contained in $\beta(Z_n)$. Of these numbers p we always choose—only in order to fix something definite—the least k (96), and, since by 108 $Z_{n'}$ is composed of Z_n and n' , we define a mapping γ of $Z_{n'}$ by requiring that for all numbers m contained in Z_n the image $\gamma(m) = \beta(m)$, and in addition that $\gamma(n') = \alpha_{n'}(k)$; this evidently similar mapping γ of $Z_{n'}$ into Σ we now regard as an image $\theta(\beta)$ of the mapping β , and in this way a mapping θ of the system Ω into itself is completely defined. After the things denoted in 126 by Ω and θ have been determined, we finally choose, for the element of Ω denoted there by ω , the given mapping α_1 ; by 126 this determines a mapping ψ of the sequence of numbers N into Ω , which, if we denote the corresponding image of an arbitrary number n not by $\psi(n)$ but by ψ_n , satisfies the conditions

$$\text{II. } \psi_1 = \alpha_1, \quad \text{III. } \psi_{n'} = \theta(\psi_n).$$

By complete induction (80) it first follows that ψ_n is a similar mapping of Z_n into Σ ; for

ρ . this is true by II for $n = 1$, and

σ . if this assertion is true for a number n , then from III and from the nature of the transition θ from β to γ described above it follows that the assertion is also true for the following number n' , as was to be proved.

Next we prove, again by complete induction (80), that, if m is any number, the special property announced above,

$$\psi_n(m) = \psi_m(m),$$

really belongs to all numbers n which are $\geq m$, and hence by 93 and 74 belong to the chain m_0 ; indeed,

ρ . this is immediately evident for $n = m$, and

σ . if this property belongs to a number n , then again from III and from the nature of θ it follows that it also belongs to the number n' , as was to be proved.

After this special property of our new series of mappings ψ_n has also been established, we can easily prove our proposition. We define a mapping χ of the sequence of numbers N by making the image $\chi(n) = \psi_n(n)$ correspond to each number n ; evidently (by 21) all mappings ψ_n are contained in this single mapping χ . Since ψ_n was a mapping of Z_n into Σ , it follows first that the sequence of numbers N is likewise mapped by χ into Σ , hence $\chi(N) \subset \Sigma$. If, further, m, n are different numbers, then by symmetry we may by 90 assume that $m < n$; then by the above $\chi(m) = \psi_m(m) = \psi_n(m)$ and $\chi(n) = \psi_n(n)$. But since ψ_n was a similar mapping of Z_n into Σ , and m, n are different elements of Z_n , $\psi_n(m)$ is different from $\psi_n(n)$, and hence also $\chi(m)$ is different from $\chi(n)$; that is, χ is a similar mapping of N . Since moreover N is an infinite system (71), the same holds by 67 for the system $\chi(N)$ similar to it, and by 68, because $\chi(N)$ is part of Σ , also for Σ , as was to be proved.

160. Proposition. A system Σ is finite or infinite according as there exists or does not exist a system Z_n similar to it.

Proof. If Σ is finite, then by 159 there are systems Z_n which cannot be similarly mapped into Σ ; since by 102 the system Z_1 consists of the single number 1 and consequently can be similarly mapped into every system, the least number k (96) to which there corresponds a system Z_k not similarly mappable into Σ must be different from 1, hence by 78 must be $= n'$, and since $n < n'$ (91), there is a similar mapping ψ of Z_n into Σ . If now $\psi(Z_n)$ were only a proper part of Σ , so that there were an element α in Σ which is not contained in $\psi(Z_n)$, then, since $Z_{n'} = \mathfrak{M}(Z_n, n')$ (108), one could extend this mapping ψ to a similar mapping ψ of $Z_{n'}$ into Σ by setting $\psi(n') = \alpha$, whereas by our assumption $Z_{n'}$ cannot be similarly mapped into Σ . Therefore $\psi(Z_n) = \Sigma$, that is, Z_n and Σ are similar systems. Conversely, if a system Σ is similar to a system Z_n , then by 119 and 67 Σ is finite, as was to be proved.

161. Definition. If Σ is a finite system, then by 160 there is one, and by 120 and 33 only one, number n to which there corresponds a system Z_n similar to the system Σ ; this number n is called the *number* of the elements contained in Σ (or also the *degree* of the system Σ), and one says that Σ consists of, or is, a system of n elements, or that the number n states how many elements are contained in Σ ¹⁷. When the numbers are used to express exactly this definite property of finite systems, they are called *cardinal numbers*. As soon as a definite similar mapping ψ of the system Z_n is chosen, by virtue of which $\psi(Z_n) = \Sigma$, to every number m contained in Z_n (that is, to every number m which is $\leq n$) there corresponds a definite element $\psi(m)$ of the system Σ , and conversely, by 26, to every element of Σ there corresponds through the inverse mapping $\bar{\psi}$ a definite number m in Z_n . Very often all elements of Σ are denoted by a single letter, for example α , to which the distinguishing numbers m are attached as subscripts, so that $\psi(m)$ is denoted by α_m . One also says that these elements have been counted and ordered in a definite way by ψ , and calls α_m the m -th element of Σ ; if $m < n$, then $\alpha_{m'}$ is called the element following α_m , and α_n is called the last element. In this counting of the elements, therefore, the numbers m again occur as ordinal numbers (73).

¹⁷For clarity and simplicity we restrict, in what follows, the concept of number entirely to finite systems; therefore, when we speak of a number of certain things, it should always already be expressed thereby that the system whose elements these things are is finite.

162. Proposition. All systems similar to a finite system possess the same number of elements.

The proof follows immediately from 33 and 161.

163. Proposition. The number of the elements contained in Z_n , that is, of those numbers which are $\leq n$, is n .

Proof. For by 32 Z_n is similar to itself.

164. Proposition. If a system consists of a single element, then the number of its elements is $= 1$, and conversely.

The proof follows immediately from 2, 26, 32, 102, and 161.

165. Proposition. If T is a proper part of a finite system Σ , then the number of the elements of T is smaller than that of the elements of Σ .

Proof. By 68, T is a finite system, hence similar to a system Z_m , where m denotes the number of the elements of T ; further, if n is the number of the elements of Σ , so that Σ is similar to Z_n , then by 35 T is similar to a proper part E of Z_n , and by 33 also Z_m and E are similar to one another. If now $n \leq m$, hence $Z_n \subset Z_m$, then by 7 E would also be a proper part of Z_m , and consequently Z_m would be an infinite system, which contradicts proposition 119; hence (by 90) $m < n$, as was to be proved.

166. Proposition. If $\Gamma = \mathfrak{M}(B, \gamma)$, where B is a system of n elements and γ denotes an element of Γ not contained in B , then Γ consists of n' elements.

Proof. For if $B = \psi(Z_n)$, where ψ denotes a similar mapping of Z_n , then by 105 and 108 this mapping can be extended to a similar mapping ψ of $Z_{n'}$ by setting $\psi(n') = \gamma$, and in fact $\psi(Z_{n'}) = \Gamma$, as was to be proved.

167. Proposition. If γ is an element of a system Γ consisting of n' elements, then n is the number of all the other elements of Γ .

Proof. For if B denotes the collection of all elements in Γ different from γ , then $\Gamma = \mathfrak{M}(B, \gamma)$; if now b is the number of the elements of the finite system B , then by the preceding proposition b' is the number of the elements of Γ , hence $= n'$, from which by 26 it follows also that $b = n$, as was to be proved.

168. Proposition. If A consists of m and B consists of n elements, and if A and B have no common element, then $\mathfrak{M}(A, B)$ consists of $m + n$ elements.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$ by 166, 164, and 135. II.

σ . If the proposition holds for a number n , then it holds also for the following number n' . In fact, if Γ is a system of n' elements, then by 167 one may put $\Gamma = \mathfrak{M}(B, \gamma)$, where γ is an element and B denotes the system of the n other elements of Γ . Now if A is a system of m elements, each of which is not contained in Γ , and hence is not contained in B either, and if one puts $\mathfrak{M}(A, B) = \Sigma$, then by our assumption $m + n$ is the number of the elements of Σ ; and since γ is not contained in Σ , by 166 the number of the elements contained in $\mathfrak{M}(\Sigma, \gamma)$ is $= (m + n)'$, hence (by 135. III) $= m + n'$. But since by 15 evidently $\mathfrak{M}(\Sigma, \gamma) = \mathfrak{M}(A, B, \gamma) = \mathfrak{M}(A, \Gamma)$, $m + n'$ is the number of the elements of $\mathfrak{M}(A, \Gamma)$, as was to be proved.

169. Proposition. If A, B are finite systems of respectively m, n elements, then $\mathfrak{M}(A, B)$ is a finite system, and the number of its elements is $\leq m + n$.

Proof. If $B \subset A$, then $\mathfrak{M}(A, B) = A$, and the number m of the elements of this system is (by 142) $< m + n$, as was asserted. If, however, B is not a part of A , and T is the system of all those elements of B which are not contained in A , then by 165 their number $p \leq n$, and since evidently

$$\mathfrak{M}(A, B) = \mathfrak{M}(A, T),$$

by 143 the number $m + p$ of the elements of this system is $\leq m + n$, as was to be proved.

170. Proposition. Every system composed of a number n of finite systems is finite.

Proof by complete induction (80). For

ρ . the proposition is self-evident by 8 for $n = 1$.

σ . If the proposition holds for a number n , and if Σ is composed of n' finite systems, let A be one of these systems and B the system composed of all the remaining ones; since their number is (by 167) $= n$, by our assumption B is a finite system. Since now evidently $\Sigma = \mathfrak{M}(A, B)$, it follows from this and from 169 that Σ is also a finite system, as was to be proved.

171. Proposition. If ψ is a non-similar mapping of a finite system Σ of n elements, then the number of the elements of the image $\psi(\Sigma)$ is smaller than n .

Proof. From all those elements of Σ which have one and the same image, choose only a single one each time, arbitrarily; then the system T of all these selected elements is evidently a proper part of Σ , because ψ is a non-similar mapping of Σ (26). At the same time, however, it is clear that the mapping of this part T contained in ψ (by 21) is similar, and that $\psi(T) = \psi(\Sigma)$. Hence the system $\psi(\Sigma)$ is similar to the proper part T of Σ , and from this our proposition follows by 162 and 165.

172. Final remark. Although it has just been proved that the number m of the elements of $\psi(\Sigma)$ is smaller than the number n of the elements of Σ , in many cases one nevertheless likes to say that the number of the elements of $\psi(\Sigma)$ is $= n$. Naturally the word number is then used in a sense different from the one used so far (161); namely, if α is an element of Σ , and a is the number of all those elements of Σ which have one and the same image $\psi(\alpha)$, then the latter, as an element of $\psi(\Sigma)$, is often still regarded as representative of a elements, which at least according to their origin may be regarded as different from one another, and accordingly is counted as an a -fold element of $\psi(\Sigma)$. In this way one arrives at the concept, very useful in many cases, of systems in which every element is furnished with a certain multiplicity number, which indicates how often it is to be counted as an element of the system. In the above case, for example, one would say that n is the number of the elements of $\psi(\Sigma)$ counted in this sense, while the number m of the really different elements of this system agrees with the number of the elements of T . Similar deviations from the original meaning of a technical term, which are nothing other than extensions of the original concepts, occur very frequently in mathematics; but it is not the purpose of this writing to go into this more closely.

Explanatory note to the preceding paper*

“What are and what should the numbers be?” became pathbreaking in two directions: for foundational research and for axiomatic set theory. Hilbert has only recently again pointed to its significance for foundational research (Math. Ann. 104); a detailed analysis of the writing by E. Zermelo is found in Landau’s obituary notice (Gött. Nachr. 1917). How strongly axiomatic set theory is influenced by Dedekind is shown by a comparison with the Zermelo axioms (Math. Ann. 65), which in part are taken directly from Dedekind’s “definitions” (§ 1 of the writing). It is known that in this connection the “axiom of infinity” had to be postulated, since Dedekind’s attempted proof (66) rests on the contradictory concept of the

*Cf. also the explanatory notes to LX and LXII.

“set of everything thinkable”; likewise, it is known that the axiom of choice enters into Dedekind’s considerations (159). Zermelo’s second proof of the well-ordering theorem can also be regarded as a transfer of the proof given here for the possibility of complete induction to transfinite induction; in the transfinite case, however, the axiom of choice already had to be added here to the other axioms implicitly used by Dedekind. Dedekind was able to avoid it for ordinary complete induction because he had at his disposal the mapping entering into the definition of the infinite. The theorem of “definition” by complete induction (126), going beyond the “proof” by complete induction, is sharply worked out for the transfinite by J. v. Neumann (Math. Ann. 99). The theorem is used in particular in the algebra of infinite domains, just as Dedekind obtains the rules of calculation for the whole numbers by means of definition by complete induction.

Noether.